



Regulation FD and the Financial Information Environment: Early Evidence

Author(s): Frank Heflin, K. R. Subramanyam and Yuan Zhang

Source: *The Accounting Review*, Jan., 2003, Vol. 78, No. 1 (Jan., 2003), pp. 1-37

Published by: American Accounting Association

Stable URL: <https://www.jstor.org/stable/3203294>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



American Accounting Association is collaborating with JSTOR to digitize, preserve and extend access to *The Accounting Review*

JSTOR

Regulation FD and the Financial Information Environment: Early Evidence

Frank Heflin

Purdue University

K. R. Subramanyam

Yuan Zhang

University of Southern California

ABSTRACT: On October 23, 2000, the SEC implemented Regulation FD (Fair Disclosure), which prohibits firms from privately disclosing value-relevant information to select securities markets professionals without simultaneously disclosing the same information to the public. We examine whether Regulation FD's prohibition of selective disclosure impairs the flow of financial information to the capital markets prior to earnings announcements. After implementation of FD, we find (1) improved informational efficiency of stock prices prior to earnings announcements, as evidenced by smaller deviations between pre- and post-announcement stock prices; (2) no reliable evidence of change in analysts' earnings forecast errors or dispersion; and (3) a substantial increase in the volume of firms' voluntary, forward-looking, earnings-related disclosures. Overall, we find no evidence Regulation FD impaired the information available to investors prior to earnings announcements, and some of our evidence is consistent with improvement.

Keywords: *disclosure; regulation; earnings announcements; capital markets; earnings forecasts; financial analysts.*

Data Availability: *All data employed in this study are commercially available from the sources described in the text.*

We are grateful to First Call/Thomson Financial for providing us with data used in this study. We thank the SEC and Financial Reporting Institute at the Leventhal School of Accounting, University of Southern California and the Krannert Graduate School of Management, Purdue University for their generous financial support. We also acknowledge Jinyoung Park's able research assistance. Finally, we thank two anonymous referees, Mark L. DeFond, William J. Kross, Jonathan Shokobin of the SEC, Frank Fernandez of the Securities Industry Association, and participants at the Financial Management Association Conference, Toronto, and research workshops at the University of Colorado at Boulder, University of Maryland, and Mellon Capital Management for their comments and suggestions. All remaining errors are our own.

*Submitted July 2001
Accepted August 2002*

I. INTRODUCTION

On October 23, 2000, the voluntary disclosure practices of corporations with publicly traded securities became subject to the requirements of the Securities and Exchange Commission's (SEC's) Regulation Fair Disclosure (FD). Regulation FD prohibits corporations from privately disclosing material information to select investors or securities markets professionals without simultaneously disclosing the same information to the public. Regulators intended FD to eliminate superior trading opportunities for recipients of firms' selective disclosures.¹ However, many securities markets professionals contend FD reduced the quantity and quality of information available to the capital markets, resulting in less accurate expectations of firm performance and larger price shocks when firms announce earnings (e.g., Securities Industry Association 2001; Association for Investment Management and Research 2001). In this paper, we empirically investigate whether the implementation of FD is associated, *on average*, with changes in the earnings-related information environment.² Specifically, we test for changes in (1) the informational efficiency of stock prices prior to firms' quarterly earnings announcements; (2) the accuracy and dispersion of analysts' earnings forecasts; and (3) the frequency of firms' voluntary disclosures. In general, we find no evidence the information available to investors prior to earnings announcements deteriorated after FD. On the contrary, we find some improvement in stock price efficiency and a substantial increase in the number of voluntary disclosures after FD's implementation.

The SEC implemented Regulation FD to address the commission's (particularly former chairman Arthur Levitt's) concern that firms communicate value-relevant information to select professional analysts and investors before disclosing it to the public.³ Selective disclosure provides select investors an informational advantage from which they can profit at the expense of others, and the SEC was concerned this practice undermines investor confidence in the integrity of the capital markets. Regulation FD now requires that, when a firm's management intentionally discloses material information to select market participants, such as securities markets professionals or investors who may trade on the information, the firm must simultaneously make public that information. When a firm's management unintentionally discloses material information to select market participants, it must make that information public as soon as practical, but no later than 24 hours after the initial disclosure.

In contrast, analysts believe direct communication between management and analysts is a primary means by which corporations communicate performance-relevant information to the capital markets. The analysts' community suggests FD has impaired information available to investors, in part because FD prohibits firms from privately guiding analysts' earnings forecasts, which form the basis for investors' earnings expectations. For example, the director of research at Thomson Financial/First Call states, "You can't give guidance to individual analysts anymore; that will inevitably lead to wider ranges in estimates and more surprises" (Williams and McGough 2000). Additionally, FD may reduce the amount of detailed, performance-relevant information firms provide analysts, which also

¹ See the U.S. Code of Federal Regulations 17 CFR Part 243.

² Although earnings-related information is not the only type of information FD addresses, it is clearly a primary target. For example, the regulation lists earnings information first among the types of information addressed and provides specific guidance on how to make a "planned disclosure of material information, such as an earnings release."

³ A number of financial press articles, such as Pulliam (1999), report selective disclosure, and the text of FD refers to several similar reports. See the U.S. Code of Federal Regulations 17 CFR Part 243.

may impair their ability to forecast earnings and make buy-sell recommendations.⁴ Furthermore, FD critics argue SEC 8-K filings and public announcements are insufficient substitutes for earnings guidance. When firms channel information through professional analysts, the analysts can guide the press, and ultimately investors, as to the information's appropriate context and meaning. However, since firms must now release information directly to the media, reporters and editors must draw inferences about context and meaning without the benefit of analysts' experience and expertise (Weber 2000). Additionally, FD critics suggest firms are less forthcoming in public announcements than in private conversations with analysts, partly because they fear litigation arising from improperly interpreted public announcements. Managers may also fear public disclosure of detailed information will benefit competitors, whereas professional analysts potentially use that information to improve earnings forecasts without disclosing competitively sensitive details (Opdyke and Lucchetti 2000; Weber 2000). Security Industry Association (SIA) spokesman Stuart Kawsell summarizes these concerns: "The playing field will be more level, but it will be empty" (Hassett 2000).

However, several factors may prevent deterioration in the information environment after the implementation of FD. First, analysts may be able to substitute information gathered from private search for information previously obtained directly from firms. For example, the majority of respondents to an Association for Investment Management and Research (AIMR) survey believe the accuracy of their earnings forecasts and the appropriateness of their stock recommendations will *not* suffer as a consequence of FD, despite reporting deterioration in private corporate communication to analysts (AIMR 2001). Second, companies may increase the quality and quantity of information dissemination through public disclosures. Finally, if private communication between firms and analysts cannot be (or is not) adequately monitored, FD may not successfully curtail selective disclosure to analysts. Therefore, the effect of FD on the financial information environment is an empirical question.

We analyze changes in various aspects of the financial information environment after the implementation of FD for a large sample of firms. Our test period consists of three post-FD quarters (the fourth quarter of 2000 and the first and second quarters of 2001). We compare each post-FD quarter to the latest like pre-FD quarter (the fourth quarter of 1999 and the first and second quarters of 2000, respectively). We first examine the effect of FD on the speed and extent to which stock prices anticipate information in upcoming earnings announcements. Specifically, we measure the "information gap" at various days prior to earnings announcements as the absolute deviation between the price on that day and the post-earnings-announcement stock price, after controlling for market-wide movements. A smaller information gap suggests the market has more information about the upcoming earnings announcement. The preponderance of our results suggests a *smaller* information gap over almost the entire quarter prior to earnings announcements, after FD. We next investigate the effect of FD on various aspects of analysts' forecasting performance. Although our univariate tests suggest that, on average, forecast accuracy declined and forecast dispersion increased after FD, regression analyses controlling for non-FD-related factors

⁴ For example, 72 percent of analysts responding to a Securities Industry Association (SIA 2001) survey and 56 percent of analysts responding to an Association for Investment Management and Research (AIMR 2001) survey indicate that the "overall quality" of information companies disseminate has declined as a result of FD. A majority of analysts responding to the AIMR survey believe the accessibility and responsiveness of corporate officers, as well as general corporate communication with analysts, have declined since FD, and 47 and 57 percent of the SIA and AIMR survey respondents, respectively, believe the companies they follow now disseminate less total information.

suggest FD had little effect on either forecast accuracy or forecast dispersion. Finally, we examine the effect of FD on the frequency of firms' voluntary public disclosures. We find a significant increase in voluntary earnings-related disclosures after FD. The increase in the number of firm-quarters with at least one disclosure is particularly significant.

In summary, we find no evidence that Regulation FD impaired the quality and quantity of investors' information prior to earnings announcements. We do not find deterioration in analysts' forecasting performance, but do find an improvement in the informational efficiency of prices prior to earnings announcements. We also find a marked increase in firms' voluntary disclosure frequency, which is consistent with firms substituting public disclosure for private communication through analysts.

Our results are potentially of interest to the SEC, which must assess the consequences of Regulation FD in a timely manner, especially given the criticism from the professional investment community. Our results are also potentially informative to the regulation's critics, largely securities markets professionals, since our systematic evidence suggests that their concerns that the regulation would have dire consequences for the financial information environment are likely unfounded. Finally, our results are potentially of interest to firms making disclosure decisions in the wake of this new regulation.

Our study provides early evidence on the cross-sectional average effects of FD on a broad range of issues related to the financial information environment. Recently, other researchers have explored the effect of FD on analysts' forecast performance in more detail. While broadly consistent with our results, this later work provides some interesting additional evidence. For example, Shane et al. (2001) find that errors in forecasts issued early in a quarter increase after FD, although there is no change for those issued later in the quarter.⁵ Mohanram and Sunder (2002) find no change in analysts' forecast accuracy or dispersion, but find an increase in the idiosyncratic component of analysts' information (but no change in the common information component).

We organize the rest of the paper as follows. Section II describes the sample. Section III discusses changes in the informational efficiency of stock prices. Section IV presents analyses of analysts' forecasts. Section V presents analyses of firms' voluntary disclosures, and Section VI concludes.

II. SAMPLE AND DATA

We assess the effect of FD by analyzing, before and after FD, various proxies for firms' financial information environment prior to quarterly earnings announcements. To align calendar and fiscal quarters, we confine our sample to December fiscal year-end firms. Not all our sample firms announced 2000 third quarter earnings before FD's effective date (October 23, 2000). Hence, we exclude the third quarter of 2000 from our analysis and define the fourth quarter of 2000 and the first and second quarters of 2001 as our post-FD quarters. We match each post-FD quarter with its most recent analogous quarter prior to October 23, 2000. Accordingly, our pre-FD quarters are the fourth quarter of 1999 and the first and second quarters of 2000. By limiting our analyses to like quarters, we control for potential differences across quarters in stock price reactions to earnings (Mendenhall and Nichols 1988), analysts' forecast accuracy and dispersion, and voluntary disclosure frequency. Comparing quarters close in calendar time minimizes the risk that changes in other economic variables contaminate our results.

⁵ Results in Shane et al. (2001) concerning errors from forecasts close to the earnings announcement are consistent with ours.

Our sample includes all December fiscal year-end firms with the following data available during our three pre- and three post-FD quarters: (1) current and previous quarters' earnings announcement dates; (2) current quarter and previous year's like-quarter actual earnings per share (EPS); (3) consensus EPS forecasts, where at least one analyst has updated his/her forecast since the previous quarter's earnings announcement; and (4) stock returns.⁶ We obtain stock return data from CRSP and earnings announcement dates and actual and forecasted EPS data from First Call. Data for control variables, which we discuss in more detail later, are obtained from First Call, CRSP, or Compustat.

For each test, if a firm enters the sample in a post-FD quarter, we require that it also enter in the corresponding pre-FD quarter. Our final sample includes 5,072 pairs of pre- and post-FD observations from 2,025 distinct firms, with 1,669, 1,715, and 1,688 firms in the fourth, first, and second quarters, respectively. Missing data on one or more control variables causes us to use fewer observations in certain tests. This selection process biases our sample toward larger firms with relatively greater amounts of information available prior to their earnings announcements (Grant 1980; Atiase 1985; Freeman 1987; Kross and Schroeder 1989). Thus, our inferences may not apply to smaller companies with smaller amounts of publicly available information. Finally, to reduce the possibility our inferences are influenced by extreme observations, we winsorize all continuous variables at the 99th percentiles of the distributions of their absolute values.

III. EVIDENCE FROM THE INFORMATIONAL EFFICIENCY OF STOCK PRICES

In an efficient market, stock prices quickly and correctly reflect all value-relevant information. Accordingly, we begin our investigation of the effect of FD on information flows to the stock markets by analyzing the speed with which pre-announcement stock prices assimilate earnings information.

Earnings announcements reveal significant information about firm performance and move stock prices (Beaver 1968). However, the stock market anticipates much of this information and most of the price movement associated with earnings news *precedes* the earnings announcement (Ball and Brown 1968; Beaver et al. 1980). Superior pre-announcement information implies the market better anticipates earnings news and moves the price closer to its full information, post-announcement level (Freeman 1987).⁷ Therefore, the absolute deviation between the price on any day prior to an earnings announcement and the post-announcement price is a measure of the information gap regarding the upcoming earnings announcement, with smaller deviations implying superior pre-announcement information.

⁶ We eliminate all firms where First Call records "DDC," which denotes a discontinuity in the EPS series arising from events such as accounting changes, mergers, and acquisitions. Also, since it corresponds to the number in the earnings announcement, we use only the earliest First Call "actual" earnings number when the database contains, due to restatements, multiple earnings numbers.

⁷ We assume the price two days after the announcement is the full information price because existing research (e.g., Morse 1981) shows prices impound most information by the day of (or after) an earnings announcement. Other research (e.g., Bernard and Thomas 1989, 1990; Freeman and Tse 1989) suggests the post-announcement price does not *fully* reflect the earnings announcement's information. We do not expect post-announcement drift to have any appreciable effect on our results because (1) the total earnings information-induced change in price occurring beyond two days after the announcement is a relatively small proportion of the total earnings-induced price change, and (2) evidence suggests the drift is due to investor naiveté regarding the magnitude of serial correlation in quarterly earnings (Ball and Bartov 1996) or researchers' difficulties specifying a quarterly earnings time-series model (Jacob et al. 2000). Neither likely has any relation to FD.

Accordingly, on each of the 64 trading days (the approximate number in a quarter) prior to the announcement we compute the absolute cumulative abnormal return as $ACAR_{i,q,x} = |\Pi_{t=-x}^{+2} (1 + AR_{i,q,t}) - 1|$ where x is the number of days the accumulation window extends *backward* from the earnings announcement date, and $AR_{i,q,t}$ is firm i 's abnormal return on day t relative to the earnings announcement date of quarter q .⁸ Abnormal returns are prediction errors from firm-specific estimations of the market model over the year ending the day before the start of the fiscal quarter.⁹ We define the earnings announcement date as the day containing the announcement if the announcement is during trading hours, and the day after if it is after trading hours.¹⁰

$ACAR_{i,q,x}$ measures the absolute percentage change in price, after abstracting from market-wide movements, from x days before to two days after an earnings announcement. Thus, it measures the quarter's earnings news that is not reflected in stock price as of x days prior to the announcement (i.e., the information gap). Higher $ACAR_{i,q,x}$'s (ACARs) indicate larger information gaps. If, as critics suggest, FD reduced the flow of earnings-related information to market participants, *ceteris paribus*, we should observe higher ACARs after than before FD. Conversely, if FD reduced the information gap, ACARs should be lower for our post- than our pre-FD quarters.

Univariate Results

Figure 1 plots ACARs from 64 trading days before to two days after earnings announcements for both the pre- and post-FD quarters. Both the mean and median post-FD ACARs, in Panels A and B, respectively, are smaller than their pre-FD counterparts in the days leading up to and including the announcement day. The distance between the pre- and post-FD ACARs widens as time extends backward from the announcement, peaking at about day -30 and remaining relatively constant back to day -64 . These plots suggest the information gap between the pre-announcement and the full-information post-announcement price is actually *smaller* post- than pre-FD.

The plots in Panels A and B display the *absolute* information flow for the pre- and post-FD quarters (i.e., they do not condition on the total information flow for the quarter). However, the post-FD plots lie below the pre-FD plots even at the beginning of the quarter, suggesting the *total* information flow over the quarter is smaller in our post- than pre-FD quarters. To measure the proportion of each quarter's total information impounded in price by any given day, we scale each day's mean or median ACAR by the mean or median, respectively, $ACAR_{i,q,-64}$, so that each plot starts the quarter at 1.0 and ends at 0.0. We

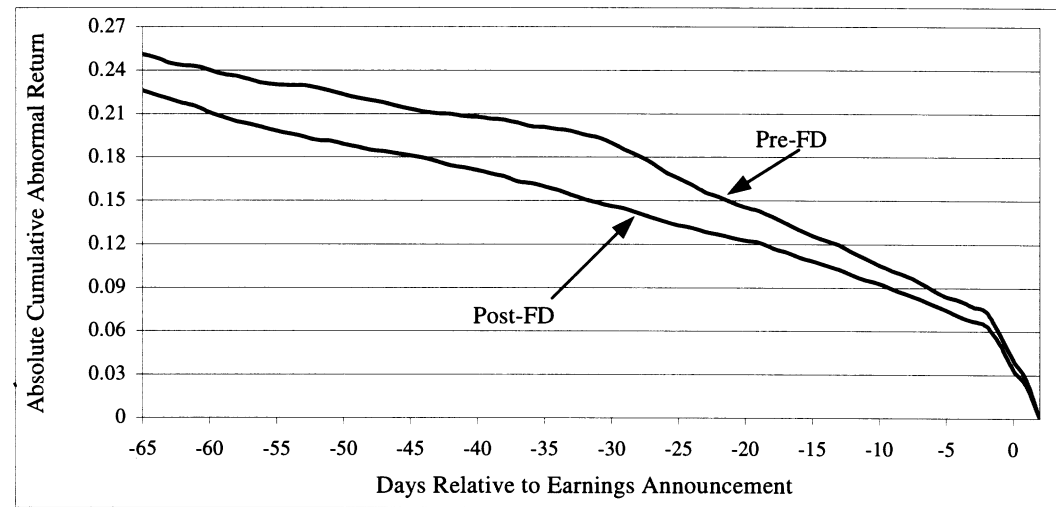
⁸ Many studies (e.g., Beaver 1968; Landsman and Maydew 2002) use cumulative return variance (i.e., cumulative squared market model prediction errors) to examine stock price reactions to earnings announcements. We use $ACAR_{i,q,x}$ because it more directly measures the information "gap" and less likely produces extreme observations. While there is little difference between the two measures in short windows, they diverge over longer windows, since daily stock returns are negatively serially correlated (Lo and MacKinlay 1988). Other studies employing absolute returns include Morse (1981), Mendenhall and Fehrs (1999), Cheon et al. (2001), and Francis et al. (2002). Nevertheless, we replicate our four-, five-, and eight-day window analyses with squared market model prediction errors, both unscaled and scaled by the market model estimation period return variance. Our inferences are unchanged.

⁹ Size-adjusted and market-adjusted returns yield qualitatively similar results.

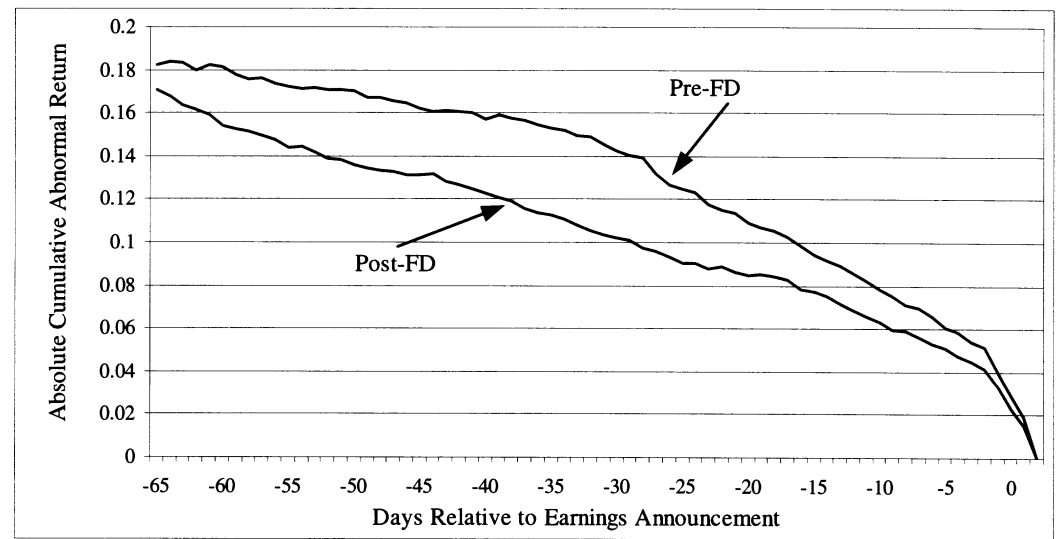
¹⁰ We were able to match all but 137 of 2,472 First Call fourth-quarter 1999 earnings announcement dates to corresponding Compustat dates. For 127 of the 137 nonmatches, we located an earnings announcement on Dow Jones or one of the wire services available through Lexis/Nexis. The earliest wire service date coincided with the First Call date 90 times, the Compustat date 18 times, and with neither 19 times. This analysis mitigates concerns about the relative accuracy of the First Call announcement dates.

FIGURE 1
Price Discovery Before and After Regulation FD^a

Panel A: Mean Absolute Cumulative Abnormal Return



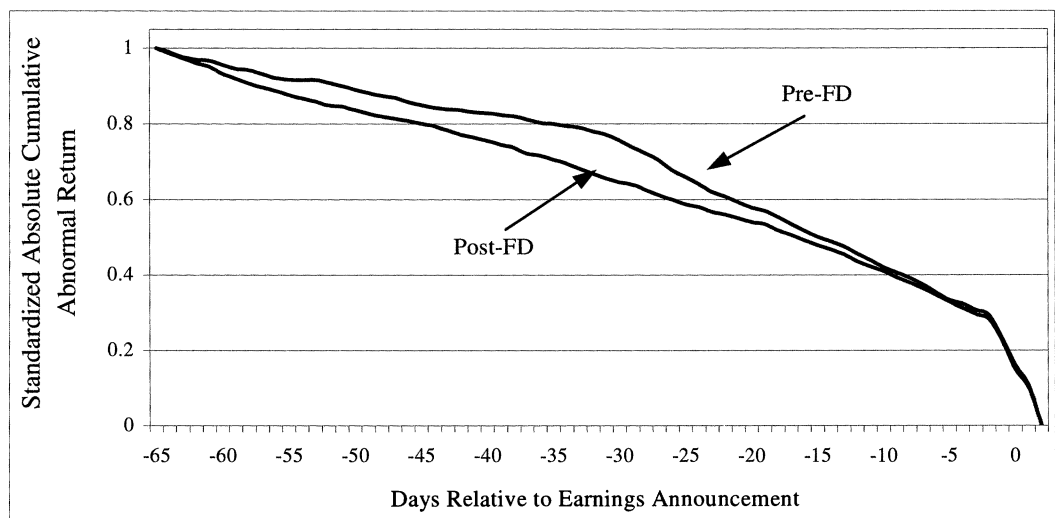
Panel B: Median Absolute Cumulative Abnormal Return



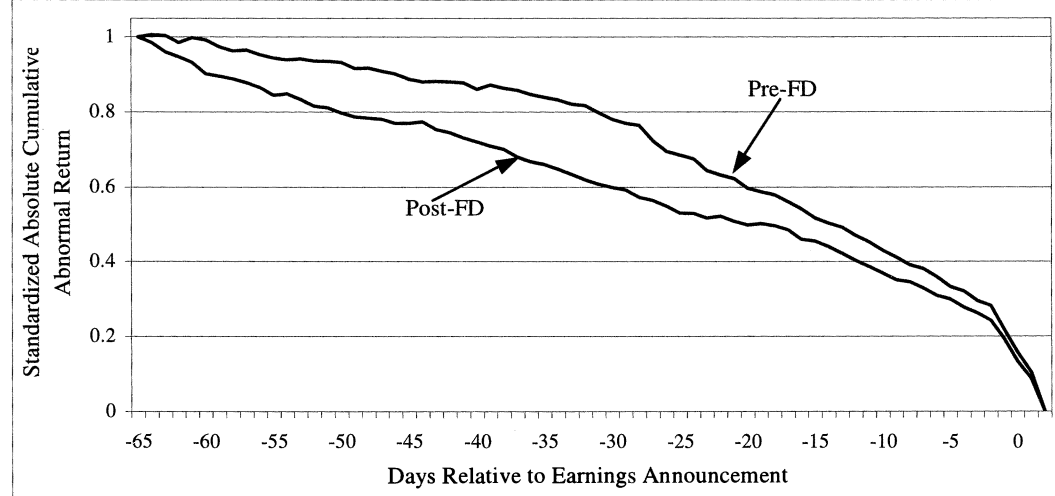
(continued on next page)

FIGURE 1 (continued)

Panel C: Daily Cross-Sectional Mean Absolute Cumulative Abnormal Return (ACAR) Scaled by Day -64 Cross-Sectional Mean ACAR



Panel D: Daily Cross-Sectional Median Absolute Cumulative Abnormal Return (ACAR) Scaled by Day -64 Cross-Sectional Median ACAR



^a Each panel depicts either absolute, or standardized absolute, cumulative abnormal returns where the accumulation period begins on day $-x$ (where $-x$ is the value on the horizontal axis) and continues through day $+2$ relative to the earnings announcement date. Abnormal returns are prediction errors from the market model, estimated over the year ending the day before the start of the relevant fiscal quarter. Each variable is winsorized at the 99th percentile of the distribution of its absolute values.

present these standardized plots in Panels C and D of Figure 1.¹¹ In general, they are consistent with the plots in Panels A and B. For the entire 67-day period, both the mean and median standardized post-FD ACARs, in Panels C and D, lie below their pre-FD counterparts, suggesting smaller information gaps post-FD and general improvement in the information available to the market.

Table 1 presents statistical tests of differences between the pre- and post-FD pooled cross-sectional mean and median ACARs for accumulation windows of five different widths. The last day of each window is the second day after the earnings announcement (i.e., day +2) and the first day is alternatively day -1, -2, -5, -10, or -30 relative to the earnings announcement. The post-FD pooled cross-sectional mean and median ACARs are lower than their pre-FD counterparts for all five windows, and all differences are statistically significant.¹² The significant differences for the (-1,+2) and (-2,+2) windows suggest that, at the time of the quarterly earnings announcement, post-FD prices are closer to their full information levels than are pre-FD prices. The differences for the longer windows suggest that, even as many as 30 days prior to the announcement, post-FD prices are closer to their full information levels than pre-FD prices. Table 1 also presents the proportion of sample firms for which ACARs either decreased or increased from the pre- to post-FD periods. The percentage of firms with lower post-FD ACARs is significantly greater than the percentage with higher post-FD ACARs for every accumulation window. In summary, standard univariate comparisons of ACARs suggest that, post-FD, superior information is available to market participants prior to earnings announcements. Further, this superiority is pervasive through the entire quarter preceding the announcement.¹³

Simple comparisons of mean or median ACARs across a small number of quarters closely aligned in time can result in erroneous inferences because prior research reports large time-series variation in the price response to earnings announcements (Mendenhall and Fehrs 1999; Landsman and Maydew 2002). Accordingly, we also assess the significance of the differences in mean (and median) ACARs between pre- and post-FD quarters by benchmarking them to the historical time-series variation in such differences. Specifically, we gather a time series of 24 overlapping seven-consecutive-quarter sets during the period from the first quarter of 1993 through the second quarter of 2000. For each of the 24 sets of seven consecutive quarters, we label the first three quarters pseudo-pre-FD quarters and the last three quarters pseudo-post-FD quarters and exclude the middle quarter.¹⁴ We then

¹¹ The standardized ACARs are conceptually appealing, but they suffer certain practical problems. First, small-denominator problems force us to scale each day's cross-sectional mean or median ACAR by the cross-sectional mean or median ACAR on day -64, rather than computing the mean or median of firm-specific scaled ACARs. Also, the scaled ACARs do not incorporate differences in informational efficiency *prior* to the current quarter.

¹² We report changes in cross-sectional means and medians, rather than mean and median within-firm changes to be consistent with our analysis in Section IV, where, as we explain later, the median within-firm change is not meaningful. However, analyses of mean and median within-firm changes from pre- to post-FD produce results very similar to the changes in means and medians reported in Table 1. Also, we do not report scaled average ACARs (as in Panels C and D of Figure 1) for brevity. However, scaled ACAR results are qualitatively similar to those reported in Table 1.

¹³ Bamber et al. (2000) report that Beaver's (1968) U-statistic is greater than one for 25 to 30 percent of a sample of firms from the 1960s. We find U-statistics greater than 1 for the (-1,+2) window for 35 percent of our sample firms prior to FD and 23 percent after, consistent with the notion that information availability improved after FD.

¹⁴ Our pseudo-pre- and pseudo-post-FD quarters are structured analogously to our actual pre- and post-FD quarters, i.e., three pre-FD quarters and three post-FD quarters separated by one excluded quarter in the middle. Also, as in our actual pre- and post-FD quarters, we require that any firm entering a pseudo-post-FD quarter must also enter its corresponding pseudo-pre-FD quarter, but we do not require that all firms have data for all pseudo-pre- and pseudo-post-FD quarters. Thus, while each pair of pseudo-pre- and corresponding pseudo-post-FD quarters employs the same firms, the set of firms used to compute the distributions for the Z-statistics changes over time.

TABLE 1
Absolute Cumulative Abnormal Returns ($ACAR_{i,q,x}$) and Pre- and Post-FD Earnings Announcements^a

	-1, +2 Window ($ACAR_{i,q,-1}$)		-2, +2 Window ($ACAR_{i,q,-2}$)		-5, +2 Window ($ACAR_{i,q,-5}$)		-10, +2 Window ($ACAR_{i,q,-10}$)		-30, +2 Window ($ACAR_{i,q,-30}$)	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median
Pre-FD ^b	0.073	0.050	0.076	0.052	0.088	0.062	0.108	0.077	0.184	0.131
Post-FD	0.064	0.041	0.068	0.044	0.079	0.053	0.095	0.064	0.152	0.104
Difference:										
Magnitude	-0.008	-0.009	-0.009	-0.009	-0.009	-0.009	-0.013	-0.013	-0.032	-0.027
p-value ^c	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Z-statistic ^d	-1.690	-2.152	-1.532	-1.775	-1.293	-1.792	-1.514	-1.935	-2.410	-2.523
p-value	(0.09)	(0.03)	(0.13)	(0.08)	(0.20)	(0.07)	(0.13)	(0.05)	(0.02)	(0.01)
% of Firms with Lower Post-FD $ACAR_{i,q,x}$	55.26		54.24		53.86		55.03		56.70	
% of Firms with Higher Post-FD $ACAR_{i,q,x}$	44.74		45.76		46.14		44.97		43.30	
p-value	(0.00)		(0.00)		(0.00)		(0.00)		(0.00)	

^a Variable definitions: $ACAR_{i,q,x}$ is the absolute cumulative abnormal return from x days before through 2 days after firm i's quarter q earnings announcement. Abnormal returns are prediction errors from the market model, estimated over the year ending the day before the start of the relevant fiscal quarter. We winsorize each $ACAR_{i,q,x}$ at the 99th percentile of the distribution of its absolute values.

^b The fourth quarter of 1999 and the first and second quarters of 2000 are pre-FD quarters, while the fourth quarter of 2000 and the first and second quarters of 2001 are post-FD quarters.

^c All p-values are two-sided. p-values for means are from t-tests of the difference between the pre- and post-FD means. For medians, p-values are from Wilcoxon two-sample tests; p-values for Z-statistics are from the student's t-distribution; p-values for differences in percentage of lower/higher post-FD ACARs are from binomial tests. There are 10,144 observations.

^d We compute mean and median ACARs for each set of pseudo-pre- and pseudo-post-FD quarters in the 24 sets of seven consecutive quarters from the first quarter of 1993 through the second quarter of 2000. We label the first three quarters in each set pseudo-pre-FD quarters and the last three pseudo-post-FD quarters. The middle quarter is discarded. Z-statistics equal $(D_x - \bar{D}_x)/\sigma_x$, where D_x is the post-FD mean or median $ACAR_{i,q,x}$ minus the pre-FD mean or median $ACAR_{i,q,x}$ and $\bar{D}_x$ and σ_x are the mean and standard deviation, respectively, of the 24 differences between the pseudo-post- and pseudo-pre-FD means or medians.

compute Z-statistics equal to $(D_x - \bar{D}_x)/\sigma_x$ where D_x is the difference between the post- and pre-FD mean ACAR and $\bar{D}_x$ and σ_x are the mean and standard deviation, respectively, of the 24 pseudo-post- and pseudo-pre-FD differences in mean ACARs. Z-statistics for differences between pre- and post-FD medians are constructed analogously.

Table 1 reveals that, for the $(-1, +2)$ window, the Z-statistic for the difference in mean ACARs is significant at the 0.09 level and for the difference in median ACARs is significant at the 0.03 level. Significance levels for the $(-2, +2)$ window are marginal ($p = 0.13$ and 0.08 for the difference in mean and median, respectively). Z-statistics for the extended windows are mixed, with significance levels ranging from 0.20 for the $(-5, +2)$ window difference in means to 0.01 for the $(-30, +2)$ window difference in medians. Collectively, these univariate time-series tests reveal no evidence that Regulation FD impaired the flow of earnings-related information and, to the extent the pre- vs. post-FD differences are significant, continue to suggest an improvement in price efficiency, albeit weakly.¹⁵

Regression Results

In this section, we identify and control for factors other than Regulation FD that may cause differences in ACARs between our pre- and post-FD periods. In general, $ACAR_{i,q,x}$ is the product of the absolute unexpected earnings (absolute surprise) as of day $-x$ and an earnings response coefficient (ERC). Holthausen and Verrecchia (1988) suggest three information-related factors that affect the ERC: inherent price variability, the precision of the market's information prior to the earnings announcement, and the precision of the earnings signal. In addition, empirical research has identified various non-information-related determinants of the ERC, such as interest rates and growth opportunities (e.g., Collins and Kothari 1989). Thus we could conceptually model ACAR as being determined by these four broad ERC determinants in addition to the (absolute) earnings surprise.

Our objective is to provide evidence on how FD affects the information available to the market prior to the earnings announcement, by observing changes in the ACARs. Accordingly, the treatment effect that we wish to capture includes any change in the ACARs due to earnings surprise attributable to FD and the effect of FD on the precision of the market's pre-announcement information. Thus, we need to control for any change in ACAR arising from earnings surprise unrelated to FD, inherent price variability, any aspect of the market's pre-announcement information precision unrelated to FD, the precision of the earnings signal, and other non-information-related determinants of the ERC. Accordingly, we identify control variables capturing these constructs. However, we do not control for the magnitude of unexpected earnings x days prior to the earnings announcement, because it is part of our treatment effect.

Our sample selection ensures each pre- and corresponding post-FD quarter contains the same firms, largely controlling for stationary firm-specific factors. We therefore focus primarily on controlling for factors affecting ACARs that likely changed from our pre- to post-FD periods but are unrelated to FD. Accordingly we estimate the following pooled cross-sectional and time-series regression:

¹⁵ Recent research reports higher absolute abnormal returns at the time of earnings announcements in the 1990s than in previous decades (Francis et al. 2001, 2002; Kim and Kross 2002; Landsman and Maydew 2002). Results in Francis et al. (2002) suggest the increase is due to an increase in disclosures concurrent with earnings announcements. However, their Table 2 shows relatively constant abnormal returns around earnings announcements in the 1990s and it is not clear the trend they document extends to our very recent sample period. Our results are consistent with the notion that FD introduced a nonstationarity by changing the information environment.

$$\begin{aligned}
 ACAR_{i,q,x} = & a_{0,x} + a_{1,x}POSTFD_q + a_{2,x}RETVAR_{i,q} \\
 & + a_{3,x}NEGCAR_{i,q} + a_{4,x}ABSCAR_{i,q} \\
 & + a_{5,x}LOSS_{i,q} + a_{6,x}NEGSPEC_{i,q} + a_{7,x}BOND30_q \\
 & + a_{8,x}EPRATIO_{i,q} + e_{i,q,x}.
 \end{aligned} \tag{1}$$

Our main variable of interest is $POSTFD_q$, which equals 1 if firm i 's earnings announcement is from our post-FD quarters, and 0 otherwise. We expect the $POSTFD_q$ coefficient to be positive if earnings-related information availability deteriorated after FD and negative if it improved.

We include $RETVAR_{i,q}$ and $NEGCAR_{i,q}$ as controls for inherent price variability. $RETVAR_{i,q}$ is the standard deviation of firm i 's abnormal returns during the market model estimation period for the relevant pre-FD quarter and controls for firm-specific inherent price variability. We expect it to be positively related to the ACARs (e.g., Beaver 1968).¹⁶ $NEGCAR_{i,q}$, which equals 1 if the cumulative abnormal return over the quarter is negative, and 0 otherwise, captures another dimension of price variability, as evidence suggests greater price movements in down markets than in up markets (Black 1976; Christie 1982).

We next include a set of variables designed to control for both the pre-announcement information precision and (absolute) earnings surprise unrelated to FD.¹⁷ We include the absolute cumulative abnormal return during the entire quarter ($ABSCAR_{i,q}$), as firm-quarters with larger total information flow are expected to have larger information gaps and larger ACARs at any given time.¹⁸ We also include $LOSS_{i,q}$ and $NEGSPEC_{i,q}$ in this category of controls. $LOSS_{i,q}$ equals 1 if firm i reports a loss in quarter q , and 0 otherwise, and $NEGSPEC_{i,q}$ is the absolute value of special items deflated by total assets, if negative, and 0 otherwise.¹⁹ We expect losses and large negative special items to impair the market's ability to forecast the upcoming earnings numbers. In addition to controlling for earnings surprise (or pre-announcement information precision), $LOSS_{i,q}$ and $NEGSPEC_{i,q}$ also capture differences in the precision of the earnings signal (Hayn 1995; Francis et al. 1996). While the

¹⁶ Because $RETVAR_{i,q}$ is constant for each pair of pre- and post-FD like quarters for each firm, its correlation with our variable of interest, $POSTFD_q$, is 0 by construction and, therefore, cannot affect $POSTFD_q$'s estimated coefficient. However, we expect $RETVAR_{i,q}$ to have considerable explanatory power for ACARs, so we include it to increase the power of our tests (Greene 1997, 403–404). Nonetheless, our inferences are unaffected if we exclude it. For brevity, we tabulate results from models excluding other (largely) time-invariant variables, although controlling for the natural log of equity market value, market model beta, and the book-to-price ratio does not change our inferences.

¹⁷ Although (absolute) surprise and pre-announcement information precision are conceptually different constructs, it is difficult to empirically parse the two. Pre-announcement information precision is an *ex ante* construct that determines the magnitude of surprise on average, while surprise measures the *ex post* realization for a particular earnings announcement.

¹⁸ Substituting the absolute value of the seasonal random-walk unexpected earnings for $ABSCAR_{i,q}$ yields qualitatively similar results.

¹⁹ We control for only negative special items because they are more common and of considerably greater magnitude than positive special items (e.g., Wild et al. 2001, 425). Furthermore, negative special items are frequently restructuring charges and asset impairments, which are both difficult to forecast. Finally, we find that the negative special-items variable we employ is more powerful (in our sample) than one indicating either positive or negative special items. Specifically, an indicator variable equaling 1 if the firm reports special items, irrespective of sign, is insignificant in all our models, in addition to having little effect on our $POSTFD_q$ coefficients.

relation of $LOSS_{i,q}$ and $NEGSPEC_{i,q}$ to multiple factors may make predicting and interpreting their coefficients more difficult, our interest is in the coefficient on $POSTFD_q$, not in the control variable coefficients.²⁰

Our remaining control variables are designed to capture non-information-related determinants of the ERC. We include $BOND30_q$, the yield on the CRSP 30-year bond index at the end of the fiscal quarter, to capture changes in interest rates because prior research (e.g., Collins and Kothari 1989) suggests price reactions to earnings announcements are decreasing in interest rates. Finally, we include $EPRATIO_{i,q}$, which is the earnings-price ratio of firm i at the end of (fiscal) quarter q . We include $EPRATIO_{i,q}$ as a proxy for expected growth in earnings (Penman 1996), since prior research (e.g., Collins and Kothari 1989) argues that growth expectations increase stock price responses to earnings.

Panels A and B of Table 2 display descriptive information on the explanatory variables in Equation (1). The average standard deviation of daily returns ($RETVAR_{i,q}$) is 0.0345. Cumulative abnormal returns are negative for just over 51 percent of our firm-quarters ($NEGCAR_{i,q}$), reflecting the general decline in market values during the 1999–2001 period. The average absolute cumulative abnormal return ($ABSCAR_{i,q}$) is 24 percent and sample firms report losses in over 19 percent of our quarters ($LOSS_{i,q}$). The average 30-year bond yield during our sample period is 5.72 percent.

Panel A of Table 2 also shows means and medians for our control variables in both our pre- and post-FD periods. The post-FD means and medians are significantly different ($p = 0.01$ or better, two-sided) from their pre-FD counterparts for all control variables (except $RETVAR_{i,q}$, which, by construction, is the same for the three post- and three pre-FD quarters).²¹ Thus, to the extent these variables are correlated with our variable of interest, $POSTFD_q$, their control is important. The correlation matrix, shown in Panel B of Table 2, reveals that $POSTFD_q$ is significantly correlated with all of our control variables, except $RETVAR_{i,q}$ ($p < 0.05$). However, with the exception of $BOND30_q$, the correlations are not economically large (less than 0.10 in magnitude). The correlation between $POSTFD_q$ and $BOND30_q$ is -0.7441 , reflecting a decrease in long-term interest rates from the fourth quarter of 1999 through the second quarter of 2001. Also, most of the control variables are modestly but statistically significantly correlated with each other.

Panel C of Table 2 presents results from estimating Equation (1) for each of the five accumulation windows. The $POSTFD_q$ coefficient is negative and significant at better than the 0.01 level (two-sided) for all five accumulation windows, suggesting that the information gap actually declines after FD, as suggested by the univariate analyses in Table 1.

We also generate Z-statistics for the $POSTFD_q$ coefficients in an analogous manner to that for the differences in mean and median ACARs in our univariate analysis. Specifically, we estimate Equation (1) for each of the 24 sets of seven consecutive quarters from the first quarter of 1993 through the second quarter of 2000. $POSTFD_q$ equals 1 for the last three quarters in each seven-quarter set, and 0 for the first three quarters. We exclude the middle quarter. The Z-statistic equals $(\hat{a}_{1,x} - \bar{a}_{1,x})/\sigma_x$, where $\hat{a}_{1,x}$ is the estimate of $a_{1,x}$ (the

²⁰ Lower earnings precision (i.e., quality) unlikely explains the drop in price reactions to earnings announcements we report in Table 1. First, in our multiple regression tests, $LOSS_{i,q}$ and $NEGSPEC_{i,q}$ at least partially control for changes in earnings precision. Second, when we regress quarterly (long-window) returns on earnings, earnings change, and a loss indicator, we find higher R^2 s after FD than before. These results indicate that earnings precision did not decline from our pre- to our post-FD period.

²¹ In all our tables, p-values associated with means are from t-tests of the difference between the means, while p-values associated with medians are from Wilcoxon two-sample tests. The Wilcoxon two-sample test can indicate a significant difference between two samples even when the sample medians are identical (e.g., if the values above the median in sample A are generally larger than the values above the median in sample B, but the values below the medians of the two samples are roughly the same).

TABLE 2
Regression of Absolute Cumulative Abnormal Returns on Post-FD Indicator and Control Variables

$$ACAR_{i,q,x} = a_{0,x} + a_{1,x}POSTFD_q + a_{2,x}RETVAR_{i,q} + a_{3,x}NEGCAR_{i,q} + a_{4,x}ABSCAR_{i,q} + a_{5,x}LOSS_{i,q} + a_{6,x}NEGSPEC_{i,q} + a_{7,x}BOND30_q + a_{8,x}EPRATIO_{i,q} + e_{i,q,x}$$

Panel A: Distributional Statistics

Control Variables ^a	Means				Medians			
	Overall	Pre-FD	Post-FD	p-value ^b	Overall	Pre-FD	Post-FD	p-value
RETVAR	0.0345	0.0345	0.0345	NA	0.0304	0.0304	0.0304	NA
NEGCAR	0.5171	0.5430	0.4913	(0.00)	1.0000	1.0000	0.0000	(0.00)
ABSCAR	0.2441	0.2548	0.2334	(0.00)	0.1723	0.1785	0.1663	(0.01)
LOSS	0.1913	0.1706	0.2120	(0.00)	0.0000	0.0000	0.0000	(0.00)
NEGSPEC	0.0036	0.0022	0.0049	(0.00)	0.0000	0.0000	0.0000	(0.00)
BOND30	5.7226	5.9718	5.4734	(0.00)	5.7089	5.8048	5.3778	(0.00)
EPRATIO	0.0145	0.0158	0.0133	(0.00)	0.0128	0.0142	0.0118	(0.00)

Panel B: Pearson Correlations^c

	RETVAR	NEGCAR	ABSCAR	LOSS	NEGSPEC	BOND30	EPRATIO
POSTFD	0.0000	-0.0518	-0.0376	0.0527	0.0854	-0.7441	-0.0954
RETVAR		0.1012	0.3084	0.4459	0.1264	0.0148	-0.3419
NEGCAR			-0.1253	0.0735	-0.0021	0.1115	-0.0852
ABSCAR				0.2146	0.0771	0.0729	-0.1421
LOSS					0.1487	-0.0273	-0.5354
NEGSPEC						-0.0252	-0.0995
BOND30							0.0410

(continued on next page)

TABLE 2 (continued)

Panel C: Regression Results

Variable ^a	(-1, +2)		(-2, +2)		(-5, +2)		(-10, +2)		(-30, +2)	
	Coefficient	p-value	Coefficient	p-value	Coefficient	p-value	Coefficient	p-value	Coefficient	p-value
Intercept	0.068	(0.00)	0.064	(0.00)	0.061	(0.00)	0.081	(0.00)	0.335	(0.00)
POSTFD	-0.012	(0.00)	-0.012	(0.00)	-0.011	(0.00)	-0.016	(0.00)	-0.053	(0.00)
RETVAR	1.207	(0.00)	1.236	(0.00)	1.304	(0.00)	1.449	(0.00)	1.475	(0.00)
NEGCAR	0.003	(0.04)	0.004	(0.02)	0.004	(0.02)	0.002	(0.31)	0.000	(0.99)
ABSCAR	0.040	(0.00)	0.051	(0.00)	0.078	(0.00)	0.120	(0.00)	0.336	(0.00)
LOSS	0.001	(0.70)	0.001	(0.65)	0.008	(0.00)	0.012	(0.00)	0.025	(0.00)
NEGSPEC	0.146	(0.00)	0.112	(0.02)	0.186	(0.00)	0.127	(0.05)	0.476	(0.00)
BOND30	-0.008	(0.01)	-0.007	(0.02)	-0.007	(0.06)	-0.009	(0.03)	-0.049	(0.00)
EPRATIO	-0.138	(0.03)	-0.107	(0.10)	-0.069	(0.34)	-0.048	(0.58)	0.010	(0.93)
R ² %	13.84		15.02		18.90		22.33		40.59	
Variable	Z-Statistic ^d	p-value	Z-Statistic	p-value	Z-Statistic	p-value	Z-Statistic	p-value	Z-Statistic	p-value
POSTFD	-2.736	(0.01)	-2.629	(0.01)	-2.108	(0.04)	-2.122	(0.03)	-3.559	(0.00)

(continued on next page)

(continued on next page)

TABLE 2 (continued)

^a Variable definitions: $ACAR_{i,q,x}$ is the absolute cumulative abnormal return from x days before through 2 days after firm i 's quarter q earnings announcement; $POSTFD_q = 1$ if the observation is from the fourth quarter of 2000 or first or second quarters of 2001, and 0 otherwise; $RETVAR_{i,q}$ is the standard deviation of firm i 's returns during the market model estimation period for the pre-FD quarter; $NEGCAR_{i,q} = 1$ if the cumulative abnormal return from 64 days before through two days after the earnings announcement is negative, and 0 otherwise; $ABSCAR_{i,q}$ is the absolute value of the cumulative abnormal return from 64 days before through two days after the earnings announcement; $LOSS_{i,q} = 1$ if firm i reports a loss in quarter q , and 0 otherwise; $NEGSPEC_{i,q}$ equals the absolute value of special items deflated by total assets if negative, and 0 otherwise; $BOND30_q$ is yield on 30-year U.S. Treasury bond as of the end of the quarter; $EPRATIO_{i,q}$ is firm i 's quarter q earnings divided by firm i 's price at the end of quarter q . Abnormal returns are prediction errors from the market model, estimated over the year ending the day before the start of the relevant fiscal quarter. We winsorize all continuous variables at the 99th percentile of the distributions of their absolute values.
^b All p -values for means are from t -tests of the difference between the pre- and post-FD means. For medians, p -values are from Wilcoxon two-sample tests. p -values for regression parameters are based on ordinary least squares standard errors. P -values for Z -statistics are from the student's t -distribution. There are 9,510 observations.
^c All correlations are significant at the 0.05 level or better (two-sided) except the correlations between (1) $NEGCAR_{i,q}$ and $NEGSPEC_{i,q}$, (2) $RETVAR_{i,q}$ and $BOND30_q$, and (3) $RETVAR_{i,q}$ and $POSTFD_q$, which are not significantly different from 0 at conventional levels.
^d We estimate the equation for each of the 24 sets of seven consecutive quarters from the first quarter of 1993 through the second quarter of 2000, with $POSTFD_q$ equal to 1 for the last three quarters in each seven-quarter set and 0 for the first three quarters. The middle quarter is discarded. The Z -statistic is equal to $(\hat{a}_{1,x} - \hat{a}_{1,x})/\sigma_x$, where $\hat{a}_{1,x}$ is the estimate of $a_{1,x}$ from the true pre- and post-FD periods and $a_{1,x}$ is the mean and σ_x the standard deviation of the 24 pseudo- $POSTFD_q$ coefficient estimates.

POSTFD_q coefficient) from our actual pre- and post-FD quarters, and $\bar{a}_{1,x}$ is the mean and σ_x the standard deviation of the 24 pseudo-POSTFD_q coefficient estimates. Panel C shows that, for all five accumulation windows, this Z-statistic is negative and statistically significant at the 0.05 level (two-sided) or better, suggesting the magnitude of the decrease in ACARs from pre- to post-FD is significantly larger than the average change in ACARs over the 1993–2000 period.

Although a number of the correlations among the control variables reported in Panel B of Table 2 are statistically significant, variance inflation factors (not tabulated) are less than 2.5, suggesting multicollinearity is inconsequential. The RETVAR_{i,q} and NEGVAR_{i,q} coefficients are positive as expected (although NEGVAR_{i,q} is not significant for the longer –10,+2 and –30,+2 windows), suggesting firms with greater past price variability and firm-quarters with negative contemporaneous returns experience greater ACARs. The positive ABSCAR_{i,q} coefficients indicate the ACARs for the various windows increase with the entire quarter's ACAR. Although not significant for every window, the LOSS_{i,q} and NEGSPEC_{i,q} coefficients are positive, suggesting that, conditional on other controls, ACARs are greater when firms report losses and negative special items. The negative BOND30_q coefficients suggest lower ACARs in quarters with higher interest rates, consistent with higher future expected rates of return (Collins and Kothari 1989). The EPRATIO_{i,q} coefficients are negative (although significant at the 0.10 level or better for the –1,+2 and –2,+2 windows only), suggesting short-window price responses are increasing in expected growth.

Our results are robust to various alternative specifications and design modifications. For example, we repeat all the univariate and regression analyses reported in Tables 1 and 2 separately for each of the three sets of like quarters in our sample period (i.e., the fourth, first, and second quarters) with generally similar results in each quarterly comparison. Our results are qualitatively similar when we estimate Equation (1) after explicitly controlling for both cross-sectional and time-series dependence.²² We also estimate Equation (1) in seasonal first-difference form, where the intercept captures the effect of FD. Again, our inferences are unchanged. Finally, although we winsorize all our continuous variables, we also estimate Equation (1) after deleting observations where Cook's D is greater than 4/N (N = the number of observations) and, separately, after deleting observations with absolute studentized residuals greater than 2, with qualitatively similar inferences. Additionally, we estimate Equation (1) after replacing ACAR_{i,q}, ABSCAR_{i,q}, and RETVAR_{i,q} with their natural logarithms, again, with qualitatively similar inferences. Thus, our inferences are unlikely driven by extreme values.

To summarize, our ACAR analyses suggest no decline in price efficiency prior to earnings announcements after the implementation of FD. On the contrary we find an *improvement* in the informational efficiency of stock prices post-FD that is systematic and statistically significant relative to random time-series variation in changes in ACARs. This improvement is also robust to controls for various determinants of ACARs. Overall, our evidence is consistent with improvement, rather than deterioration in firms' financial information environments post-FD.

²² Since standard errors for the Z-statistics are from the time-series of parameter estimates, our Z-statistics are not subject to the influence of cross-sectional correlation (Bernard 1987). To control for serial dependence, we estimated Equation (1) using the first differences of the firm-specific pre- and post-FD averages (across the three quarters). In this specification, where the intercept captures the effect of FD, there is only one observation per firm. Finally, to control simultaneously for both cross-sectional and time-series dependence we estimate Equation (1) on the subset of firms with complete data in all three pre- and three post-FD quarters, using a generalized least squares procedure as described in Parks (1967). Our results are qualitatively unchanged under all these alternative specifications.

IV. EVIDENCE FROM ANALYSTS' FORECASTS

In this section we investigate the effect of FD on sell-side analysts' forecasts. The SEC issued FD to prevent managers from privately communicating material new information to securities professionals, such as analysts. Critics of FD suggest this will impair the accuracy of analysts' forecasts. However, analysts could respond to reduced information flows from managers by increasing their private information search, and firms may respond to additional limits on private communication by increasing public disclosures. Therefore, the effect of FD on analysts' performance is an empirical question.

Univariate Results

We consider two traditional measures of analysts' forecast performance: forecast error and dispersion. Following Brown et al. (1987), Ziebart (1990), Ajinkya et al. (1991), Lang and Lundholm (1996), and others, we measure forecast error as the absolute value of the difference between actual earnings per share and the mean analyst forecast, both as reported by First Call, as of the date of the most recently updated forecast prior to the earnings announcement. We measure forecast dispersion as the standard deviation of individual analysts' forecasts for a given firm, as of the date of the last forecast of quarterly earnings prior to the earnings announcement, since our objective is to examine changes in the information environment near the date of the earnings announcement. We scale both forecast error and dispersion by price at the end of the *control* (i.e., *pre-FD*) quarter, to prevent changes in our measures simply because of changing stock prices. Using price from the same day to deflate both our pre- and post-FD measures controls for cross-sectional scale differences without contaminating the results with deflator changes (Jacob et al. 1999). Finally, computing forecast dispersion requires that at least two analysts issue forecasts of the quarter's earnings. Thus, our analyses of forecast dispersion employ fewer observations than our analyses of forecast errors.

Table 3 reveals that mean and median absolute consensus forecast errors are significantly higher ($p = 0.02$ or better, two-sided) in the post-FD period than in the pre-FD period. Table 3 also presents Z-statistics for the significance of the differences in mean and median absolute consensus forecast errors, relative to their time-series variation, which we compute using a procedure analogous to that described in Section III. The Z-statistic for the difference in means is significant ($p = 0.02$, two-sided), although the Z-statistic is not significant for the difference in medians. However, more firms' absolute forecast errors increase than decrease after FD ($p < 0.01$, two-sided).

Table 3 also reveals that mean and median forecast dispersion are significantly higher post-FD, although the Z-statistic is significant only for the change in means. The proportion of firms with increased forecast dispersion significantly exceeds that with decreased dispersion ($p < 0.01$, two-sided). In summary, our univariate results provide some evidence that FD impaired analysts' forecast accuracy and increased dispersion.²³

²³ Due, in part, to First Call's rounding of both actual and forecasted EPS to two decimal places, almost a fifth of the forecast errors are zero in both the pre- and post-FD periods. In cases where the forecast error for a firm is zero in both our pre- and matched post-FD quarters, the change in absolute forecast error is also zero. While the change in absolute forecast error is often positive or negative, these zero changes lie in the middle of the distribution, making the median of the within-firm changes in absolute forecast errors exactly zero. The median absolute forecast errors during both the pre- and the post-FD periods, however, are nonzero because the zero forecast errors cluster at the left tail of the distribution of forecast errors. Therefore, test results in Table 3 are based on changes in cross-sectional medians, and to maintain consistency, changes in cross-sectional means, rather than medians and means of the within-firm changes. However, tests based on means of the within-firm pre- to post-FD differences yield inferences similar to those drawn from Table 3.

TABLE 3
Absolute Analyst Forecast Error and Dispersion in Analysts' Forecasts Before and After FD^a

	Absolute Forecast Error (n = 10,144)		Forecast Dispersion (n = 8,222)	
	Mean	Median	Mean	Median
Pre-FD	3.176 ^b	1.049	1.353	0.588
Post-FD	4.096	1.127	1.596	0.682
Difference:				
Magnitude	0.920	0.078	0.243	0.094
p-value ^c	(0.00)	(0.02)	(0.00)	(0.00)
Z-Statistic ^d	2.350	0.579	1.981	1.349
p-value	(0.02)	(0.56)	(0.05)	(0.18)
% of Firm-Quarters with Smaller Forecast Metrics Post-FD	43.20		40.87	
% of Firm-Quarters with Larger Forecast Metrics Post-FD	45.17		55.85	
p-value	(0.00)		(0.00)	

^a Variable definitions: Absolute forecast error is the absolute value of the difference between actual earnings and the mean of the individual analyst forecasts, both as reported by First Call. Dispersion is the standard deviation of the individual analysts' forecasts. We compute forecast dispersion using only firm-quarters where at least two analysts follow the firm. We scale absolute analyst forecast error and dispersion, for each pre-FD and corresponding post-FD quarter, by share price at the end of the pre-FD quarter, and winsorize scaled forecast error and dispersion at the 99th percentile of the distributions of their absolute values.

^b For presentation purposes, we multiplied all means and medians in this table by 1,000.

^c All p-values are two-sided. p-values for means are from t-tests of the difference between the pre- and post-FD means. For medians, p-values are from Wilcoxon two-sample tests. p-values for Z-statistics are from the student's t-distribution. p-values for difference in percentages are from binomial tests.

^d We compute means and medians for each set of pseudo-pre- and pseudo-post-FD quarters in the 24 sets of seven consecutive quarters from the first quarter of 1993 through the second quarter of 2000. We label the first three quarters in each set pseudo-pre-FD quarters and the last three pseudo-post-FD quarters. The middle quarter is discarded. Z-statistics equal $(D - \overline{D})/\sigma$, where D is the post-FD mean or median forecast metric (i.e., absolute forecast error or dispersion) minus the pre-FD mean or median absolute forecast metric, and $\overline{D}$ and σ are the mean and standard deviation, respectively, of the 24 differences between the pseudo-post- and pseudo-pre-FD means or medians.

Regression Results

In this section we control for changes in the economic environment unrelated to FD that could be correlated with analysts' forecast performance. Specifically, we estimate the following model:

Forecast Metric_{i,q} = b₀ + b₁POSTFD_q + b₂SRWABSUE_{i,q} + b₃NEGUE_{i,q}
+ b₄LOSS_{i,q} + b₅NEGSPEC_{i,q} + b₆GDPSHOCK_q
+ b₇DAYS_{i,q} + b₈LAGMETRIC_{i,q} + e_{i,q}

(2)

where Forecast Metric_{i,q} is either the mean absolute forecast error or the standard deviation

of individual analysts' forecasts. Again, our matched sample largely controls for stationary firm-specific factors. We therefore focus on controlling for factors affecting analysts' forecasting accuracy that likely changed from our pre- to post-FD periods, but are unrelated to FD.

We first include as controls several earnings characteristics that may affect forecasting accuracy. Since large changes in earnings are more difficult to forecast than small changes (e.g., Kross et al. 1990; Lang and Lundholm 1996), we include absolute seasonal random-walk unexpected earnings, deflated by pre-FD quarter-end stock price ($SRWABSUE_{i,q}$). Further, declining or negative earnings and earnings with negative special items may contain transitory components that analysts have difficulty anticipating. Therefore, we include the indicator variables $NEGUE_{i,q}$, which equals 1 if firm i 's quarter q earnings are below earnings from the same quarter of the previous year, and 0 otherwise, and $LOSS_{i,q}$, which equals 1 if firm i 's quarter q earnings are negative, and 0 otherwise. We also include $NEGSPEC_{i,q}$, which is defined as in Equation (1). Since analysts more likely experience difficulty forecasting earnings in periods of substantial economic change, we include the surprise in gross domestic product ($GDPSHOCK_q$), which we measure as the absolute value of the quarterly change in the seasonal growth rate in gross domestic product.²⁴ We include the log of the average number of days by which the forecast precedes the earnings announcement ($DAYS_{i,q}$), since later forecasts are more accurate than earlier forecasts (Kross et al. 1990; Clement 1999). Finally, since these variables likely do not fully capture firm-specific variation in our forecast metrics, we include each firm's forecast metric from the same quarter the year prior to our pre-FD quarters (i.e., $LAGMETRIC_{i,q} = LAGAFE_{i,q}$ or $LAGDISP_{i,q}$).²⁵

Panel A of Table 4 presents descriptive statistics regarding our control variables. The mean of $SRWABSUE_{i,q}$ indicates that the current quarter's earnings differs, on average, from earnings four quarters prior by 1.1 percent of stock price, and the mean of $NEGUE_{i,q}$ indicates that the current quarter's earnings is below the same quarter's earnings from the previous year in 36.76 percent of our firm-quarters. During our sample period, the seasonal growth rate in gross domestic product changed by an average of about 0.95 percent. On average, 19.47 days (natural log is 2.969) elapse between the last forecast prior to the earnings announcement and the announcement. Over the three quarters 1 year prior to our three pre-FD quarters, analyst forecast errors averaged 0.3 percent of stock price and the average within-firm standard deviation of forecasts was 0.0013.

Panel A of Table 4 also presents pre- and post-FD means and medians for our control variables. The pre-FD means and medians are significantly different from the post-FD means and medians for all control variables (except $LAGAFE_{i,q}$ and $LAGDISP_{i,q}$, the two $LAGMETRIC_{i,q}$ variables, which, by construction, do not vary from our pre- to post-FD

²⁴ We obtained gross domestic product data from the Federal Reserve Bank of St. Louis web site: <http://research.stlouisfed.org/fred/data/gdp.html>.

²⁵ $LAGMETRIC_{i,q}$ differs for each of the three pre-FD quarters for each firm and these three values are repeated for the post-FD quarters. Since it does not vary from the pre- to the post-FD quarters, excluding it will not bias the coefficient on $POSTFD_q$, but will reduce the power of our tests and can, therefore, affect our inferences. We include it because we expect it to be an important explanator of forecast error and dispersion and we wish to minimize the risk of failing to reject the hypothesis that FD had no impact. We also estimate Equation (2) including several other cross-sectional variables suggested by prior research, including the firm's number of business segments, the firm's book-to-price ratio, institutional ownership, and the number of analysts following the firm (Dempsey 1989; Kross et al. 1990; Clement 1999; Chaney et al. 1999; Lim 2001). We also included the natural log of the firm's equity market value (Lang and Lundholm 1996) and the number of years since the firm listed on an exchange. Inclusion of these variables does not affect our inferences regarding the $POSTFD_q$ coefficient.

TABLE 4
Regression of Absolute Analyst Forecast Error or Dispersion on Post-FD Indicator and Control Variables

$$\text{Forecast Metric}_{i,q} = b_0 + b_1\text{POSTFD}_q + b_2\text{SRWABSUE}_{i,q} + b_3\text{NEGUE}_{i,q} + b_4\text{LOSS}_{i,q} + b_5\text{NEGSPEC}_{i,q} + b_6\text{GDPSHOCK}_q + b_7\text{DAYS}_{i,q} + b_8\text{LAGMETRIC}_i + e_{i,q}$$

Panel A: Distributional Statistics

Control Variables ^a	Means			p-value ^b	Medians			p-value
	Overall	Pre-FD	Post-FD		Overall	Pre-FD	Post-FD	
SRWABSUE	0.0110	0.0099	0.0120	(0.00)	0.0042	0.0039	0.0047	(0.00)
NEGUE	0.3676	0.2932	0.4421	(0.00)	0.0000	0.0000	0.0000	(0.00)
LOSS	0.1759	0.1572	0.1946	(0.00)	0.0000	0.0000	0.0000	(0.00)
NEGSPEC	0.0035	0.0024	0.0047	(0.00)	0.0000	0.0000	0.0000	(0.00)
GDPSHOCK	0.9546	0.5601	1.3492	(0.00)	0.6200	0.3960	1.6030	(0.00)
DAYS	2.9690	2.9248	3.0132	(0.00)	3.0910	3.0910	3.1355	(0.00)
LAGAFE	0.0030	0.0030	0.0030	NA	0.0009	0.0009	0.0009	NA
LAGDISP	0.0013	0.0013	0.0013	NA	0.0005	0.0005	0.0005	NA

Panel B: Correlations^c

	SRWABSUE	NEGUE	LOSS	NEGSPEC	GDPSHOCK	DAYS	LAGAFE	LAGDISP
POSTFD	0.0565	0.1544	0.0491	0.0771	0.5940	0.0377	0.0000	0.0000
SRWABSUE		0.1476	0.2642	0.0865	0.0581	0.0305	0.3175	0.2768
NEGUE			0.3103	0.1111	0.0769	0.0015	0.0121	0.0042
LOSS				0.1410	0.0118	0.0573	0.2250	0.2359
NEGSPEC					0.0856	0.0225	0.0222	0.0170
GDPSHOCK						0.0722	−0.0203	−0.0231
DAYS							0.0582	0.0103
LAGAFE								0.4169

(continued on next page)

TABLE 4 (continued)

Panel C: Regression Results		Absolute Forecast Error (n = 8,180)		Forecast Dispersion (n = 6,850)	
Variable		Coefficient	p-value	Coefficient	p-value
Intercept		-0.001	(0.00)	0.000	(0.00)
POSTFD ^d		0.246	(0.12)	-0.068	(0.24)
SRWABSUE		0.216	(0.00)	0.072	(0.00)
NEGUE		0.000	(0.05)	0.000	(0.00)
LOSS		0.003	(0.00)	0.001	(0.00)
NEGSPEC		0.024	(0.00)	0.004	(0.03)
GDPSHOCK		-0.000	(0.50)	0.000	(0.00)
DAYS		0.001	(0.00)	-0.000	(0.39)
LAGMETRIC		0.031	(0.00)	0.156	(0.00)
R ² %	40.65				36.95
Variable		Z-Statistic ^e	p-value	Z-Statistic	p-value
POSTFD		1.082	(0.28)	1.219	(0.22)

(continued on next page)

TABLE 4 (continued)

^a Variable definitions: POSTFD _q = 1 if the observation is from the fourth quarter of 2000 or first or second quarters of 2001, and 0 otherwise; SRWABSUE _{i,q} = absolute seasonal random-walk unexpected earnings; NEGUE _{i,q} = 1 if earnings are below the same quarter's earnings from the previous year, and 0 otherwise; LOSS _{i,q} = 1 if firm <i>i</i> reports a loss in quarter <i>q</i> , and 0 otherwise; NEGSPPEC _{i,q} equals the absolute value of special items deflated by total assets if the firm reports negative special items, and 0 otherwise; GDPSHOCK _q is the absolute value of the quarterly change in the seasonal growth rate in gross domestic product; DAYS _{i,q} = the log of the number of days the forecast precedes the earnings announcement; LAGAFE _{i,q} = firm <i>i</i> 's absolute forecast error from the same quarter the year prior to pre-FD quarter <i>q</i> ; LAGDISP _{i,q} = firm <i>i</i> 's forecast dispersion from the same quarter the year prior to pre-FD quarter <i>q</i> ; LAGMETRIC _{i,q} = either LAGAFE _{i,q} or LAGDISP _{i,q} ; Absolute forecast error is the difference between actual earnings and the mean of the individual analyst forecasts. Forecast dispersion is the standard deviation of the individual analysts' forecasts. Absolute forecast error, forecast dispersion, and SRWABSUE _{i,q} for each pre-FD and corresponding post-FD quarter, are divided by share price at the end of the pre-FD quarter. We winsorize all continuous variables at the 99th percentile of the distributions of their absolute values.
^b All <i>p</i> -values for regression parameters are based on ordinary least squares standard errors. <i>p</i> -values for <i>Z</i> -statistics are from the student's <i>t</i> -distribution.
^c All correlations are significant at the 0.05 level (two-sided) or better except for the correlation between GDPSHOCK _q and LAGAFE _{i,q} , which is significant at the 0.06 level (two-sided), and the correlations between GDPSHOCK _q and LOSS _{i,q} , NEGUE _{i,q} and LAGAFE _{i,q} , NEGUE _{i,q} and LAGDISP _{i,q} , and LAGDISP _{i,q} and LAGDISP _{i,q} , which are not significant at conventional levels.
^d For presentation purposes, we multiplied this coefficient by 1,000.
^e We estimate the equation for each of the 24 sets of seven consecutive quarters from the first quarter of 1993 through the second quarter of 2000, with POSTFD _q equal to 1 for the last three quarters in each seven-quarter set and 0 for the first three quarters. The middle quarter is discarded. The <i>Z</i> -statistic is equal to $(b_1 - \bar{b}_1)/\sigma_{b_1}$, where $\bar{b}_1$ is the estimate of b_1 from the true pre- and post-FD periods, and $\bar{b}_1$ is the mean and σ_{b_1} the standard deviation of the 24 pseudo-POSTFD _q coefficient estimates.

quarters). In particular, the difference between the pre- and post-FD means for $SRWABSUE_{i,q}$, $NEGUE_{i,q}$, $LOSS_{i,q}$, and $NEGSPEC_{i,q}$ suggest that, in our post-FD period, earnings had more time-series variability, were more often below four-quarter-ago earnings, were more likely negative, and contained more negative special items. Thus, to the extent these factors impair analysts' forecasting ability, it is important to control for their effects.

Panel B of Table 4 shows that the variables most correlated with $POSTFD_q$ are $GDPSHOCK_q$ and $NEGUE_{i,q}$, although $POSTFD_q$ is significantly correlated with all the control variables (except $LAGAFE_{i,q}$ and $LAGDISP_{i,q}$, which, by construction, do not vary from our pre- to post-FD quarters). Again, most of the correlations among the control variables are small in magnitude, but statistically significant ($p = 0.05$, two-sided).

We present results from estimating Equation (2) with absolute forecast errors as the dependent variable in Panel C of Table 4. The coefficient on $POSTFD_q$ is positive but significant at only the 0.12 level (two-sided). Further, the Z-statistic for the $POSTFD_q$ coefficient, which we construct analogously to the Z-statistics for Equation (1), is insignificant ($p = 0.28$, two-sided). We also estimate Equation (2) with forecast dispersion as the dependent variable. In this estimation, the $POSTFD_q$ coefficient is negative, but not statistically significant ($p = 0.24$, two-sided). Its Z-statistic is positive with a p-value (two-sided) of 0.22. Thus, our regression results suggest that, after controlling for various characteristics of the forecasting environment, there is no discernable change in the accuracy of analysts' forecasts or in their dispersion after FD's implementation.²⁶

Coefficients on control variables suggest analysts have more trouble forecasting firm-quarters (1) with greater time-series forecast errors ($SRWABSUE_{i,q}$), (2) with declining earnings ($NEGUE_{i,q}$), (3) where firms report losses ($LOSS_{i,q}$), and (4) where firms report negative special items ($NEGSPEC_{i,q}$). Their forecasts are less accurate the longer the forecast horizon ($DAYS_{i,q}$) and they are able to reach less consensus in periods of significant economic change ($GDPSHOCK_q$). Past forecast error and past consensus ($LAGMETRIC_{i,q}$) are positively associated with current forecast error and current consensus, respectively. Despite the correlations mentioned above, none of the variance inflation factors suggest the presence of multicollinearity.

We repeat the analyses reported in Tables 3 and 4 separately for each of the three sets of like quarters in our sample period (i.e., the fourth, first, and second quarters), with generally similar results in each quarterly comparison. We estimate Equation (2) in seasonal first-difference form. Our inferences are unchanged. We also obtain qualitatively similar results when we estimate Equation (2) after explicitly controlling for both cross-sectional and time-series dependence, in similar manner to that in the case of the stock returns' analyses. Also, our estimations of Equation (2) after deleting observations where Cook's D is greater than $4/N$ (N = the number of observations) and, separately, after deleting observations with absolute studentized residuals greater than 2 do not change our inferences. Our results are therefore unlikely driven by outlier effects.

In conclusion, our univariate evidence weakly suggests an increase in analysts' forecast errors and dispersion after FD. However, our regression analyses controlling for other likely determinants of analysts' forecast errors and dispersion (e.g., earnings variability, earnings declines, losses, negative special items, general economic changes, forecast timing, and other firm-specific factors) suggest no change in forecast errors or dispersion after FD. Thus, we conclude that the evidence does not support the contention that FD impaired analysts' forecasting ability.

²⁶ When we estimate Equation (2) in each size (defined as market value of equity) quartile, the results suggest analyst forecast errors *increased* post-FD for the *smallest* quartile and forecast dispersion *decreased* for the *largest* quartile. We leave further exploration of the differential impact of FD across firms to future research.

V. EVIDENCE FROM VOLUNTARY EARNINGS-RELATED DISCLOSURES

In principle, FD should prevent managers from privately communicating value-relevant information to select analysts. However, we find no deterioration in either the informational efficiency of stock prices or analysts' forecasting performance post-FD. We investigate whether firms are providing information through alternative channels, such as public disclosure.²⁷ This explanation is consistent with prior research suggesting firms are more likely to issue voluntary disclosures in the presence of greater information asymmetry (e.g., Ajinkya and Gift 1984; Diamond and Verrecchia 1991; Lang and Lundholm 1993). Accordingly, we examine whether the frequency of voluntary earnings-related disclosures increases after FD.

Univariate Results

Our voluntary disclosure data are from First Call's "Corporate Issued Guidance" database. We restrict our analyses to disclosures First Call classifies as "earnings-related." Our data include managers' forecasts of current period's and future periods' earnings, both quarterly and annual (although a few also pertain to revenue forecasts). We include point and range estimates, upper and lower bounds, and qualitative disclosures. We include all disclosures from the day after the previous quarter's earnings announcement to the day of the current quarter's earnings announcement for each of our six (three pre- and three post-FD) quarters.

Panel A of Table 5 presents descriptive information about the disclosures in our sample, both before and after FD. These disclosures more than doubled, from 1,160 in our pre-FD period to 2,981 in our post-FD period, consistent with firms responding to FD by increasing public disclosures. The change in composition is particularly striking. Prior to FD, approximately one-third (31.55 percent) of these disclosures were point estimates, where management provided a specific estimate, and approximately one-third (35.52 percent) were range estimates, where management provided a forecast of the range in which future earnings would fall. The remaining third included upper or lower bounds, purely qualitative descriptions, or other earnings-related information. After FD, however, nearly two-thirds (62.83 percent) of the voluntary earnings-related disclosures are range estimates, and less than one-quarter (21.67 percent) are point estimates. Thus, our data suggest that, while firms issued more forecasts of each specificity level after FD, they issued relatively more range and relatively fewer point estimates after FD than before.

Panel B of Table 5 presents the average number of voluntary earnings-related management disclosures per firm-quarter during the pre- and post-FD periods and also presents statistical tests of differences. The panel categorizes the disclosures into those about current (upcoming) vs. future earnings. We classify disclosures as about current earnings if the disclosure concerns the upcoming earnings report. We classify disclosures as about future earnings if the disclosure concerns future periods' earnings. The mean number of voluntary, earnings-related management disclosures per firm-quarter is less than 1, and the medians (not tabulated) are 0. However, the mean number of disclosures per firm-quarter increased significantly after FD, from 0.229 to 0.588. The mean number of disclosures about current earnings nearly doubled from 0.132 to 0.236, and the mean number of disclosures about future earnings more than tripled, from 0.097 to 0.352, after FD. All three differences are significant at better than the 0.01 level. The Z-statistics, computed analogously to those presented in earlier sections are significant at the 0.01 level, implying that the magnitudes of these increases in voluntary, earnings-related disclosures are unusual compared to their

²⁷ Another explanation is that analysts increased their private search activity in response to reduced information flows. Mohanram and Sunder (2002) find evidence consistent with this hypothesis.

TABLE 5
Voluntary Forward-Looking Earnings-Related Disclosures Firms Issued Before and After FD

Panel A: Types of Voluntary Disclosures

	Type of Disclosure										
	Point Estimates		Range Estimates		Upper or Lower Bounds Only		Qualitative		Other Earnings Related		
	Number	Percent	Number	Percent	Number	Percent	Number	Percent	Number	Percent	
Pre-FD	366	31.55	412	35.52	119	10.26	255	21.98	8	0.69	1,160
Post-FD	646	21.67	1,873	62.83	169	5.67	258	8.65	35	1.17	2,981
Change	290	-9.88	1,461	27.31	50	-4.59	3	-13.33	27	0.48	1,831

Panel B: Current and Future Quarters' Earnings-Related Disclosures

	Mean Disclosures per Firm-Quarter: All Firm-Quarters ^a				Proportion of Firm-Quarters with at Least One Disclosure				Mean Disclosures per Firm-Quarter: Only Disclosing Firms ^c			
	Current Earnings ^b		Future Earnings		Current Earnings		Future Earnings		Current Earnings		Future Earnings	
	Number	Percent	Number	Percent	Number	Percent	Number	Percent	Number	Percent	Number	Percent
Pre-FD	0.132	0.097	0.229	0.132	11.24%	7.20%	15.04%	0.334	0.244	0.577	0.334	0.244
Post-FD	0.236	0.352	0.588	0.236	20.15%	21.27%	32.35%	0.595	0.889	1.484	0.595	0.889
Difference:												
Magnitude	0.104	0.256	0.359	0.104	8.91%	14.07%	17.31%	0.261	0.645	0.906	0.261	0.645
p-value ^d	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Z-Statistic ^e	2.862	8.741	5.746	2.862	2.377	7.150	4.693	1.402	5.202	3.231	1.402	5.202
p-value	(0.00)	(0.00)	(0.00)	(0.00)	(0.02)	(0.00)	(0.00)	(0.16)	(0.00)	(0.00)	(0.00)	(0.00)
By Firm Differences:												
% of Firm-quarters with Fewer Disclosures Post-FD	7.69	4.51	8.62	7.69	7.26	4.02	7.26					
% of Firm-quarters with More Disclosures Post-FD	16.74	19.36	27.52	16.74	16.17	18.10	24.57					
p-value	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)					

(continued on next page)

TABLE 5 (continued)

^a Mean Disclosures per Firm-Quarter: All Firm-Quarters is the total number of disclosures in a quarter, by all firms, divided by the number of sample firms, calculated separately for the pre- and post-FD periods.
^b "Current Earnings" columns refer to voluntary disclosures about the upcoming earnings announcements. "Future Earnings" columns refer to voluntary disclosures about earnings for future periods. The "Total" column includes both types.
^c Mean Disclosures per Firm-Quarter: Only Disclosing Firms is the total number of disclosures in a quarter, by just those firms making at least one disclosure in either the pre- or post-FD periods, divided by the total number of sample firms making at least one disclosure in either the pre- or post-FD periods.
^d All p-values are two-sided. p-values for means are from t-tests of the difference between the pre- and post-FD means. For medians, p-values are from Wilcoxon two-sample tests. p-values for Z-statistics are from the student's t-distribution. p-values for difference in percentages are from binomial tests. There are 10,144 observations.
^e We compute means and medians for each set of pseudo-pre- and pseudo-post-FD quarters in the 16 sets of seven consecutive quarters from the first quarter of 1995 through the second quarter of 2000. We label the first three quarters in each set pseudo-pre-FD quarters and the last three pseudo-post-FD quarters. The middle quarter is discarded. Z-statistics equal $(D - \bar{D})/\sigma$, where D is the post-FD mean minus the pre-FD mean, and $\bar{D}$ and σ are the mean and standard deviation, respectively, of the 16 differences between the pseudo-post- and pseudo-pre-FD means.

historical variation.²⁸ Finally, for both current and future earnings, significantly more firms increased than decreased their disclosures after FD.

The middle three columns in Panel B of Table 5 show that the proportion of firm-quarters containing at least one voluntary disclosure increases significantly from the pre- to the post-FD period. Across the three pre-FD quarters, only 15.04 percent of the firm-quarters contained at least one disclosure. After FD, firms increased disclosures about both current and future earnings such that 32.35 percent of the firm-quarters contained at least one disclosure.²⁹ The time-series Z-statistics are also positive and significant at better than the 0.02 level (two-sided). Overall, we find a marked increase in the number of firm-quarters with at least one voluntary, earnings-related disclosure after FD.

In the rightmost three columns, we track the mean number of disclosures per firm-quarter for only those firms making at least one voluntary disclosure in either our pre- or post-FD periods.³⁰ These columns reveal that disclosing firms increased their disclosure frequency after FD. The average number of disclosures per firm-quarter more than doubled from 0.577 prior to FD to 1.484 after FD. This increase is evident in disclosures about both current and future earnings. The average number of disclosures, per firm-quarter, about current earnings increased from 0.334 to 0.595, and the average number of disclosures, per firm-quarter, about future earnings more than tripled from 0.244 to 0.889. All are statistically significant, in standard tests, at the 0.01 level (two-sided) or better. The time-series Z-statistics are significant, however, only for disclosures about future earnings and for total disclosures.³¹

Regression Analyses

In this section, we report results from estimating an ordered logit model that controls for non-FD factors influencing managers' voluntary disclosure decisions. Again, we focus on factors that could have changed between the pre- and post-FD periods, rather than purely cross-sectional variables. The model we estimate is:

$$\begin{aligned} \text{DISC}_{i,q} = & c_0 + c_1 + c_2\text{POSTFD}_q + c_3\text{ABSCAR}_{i,q} + c_4\text{EPDEV}_{i,q} \\ & + c_5\text{EARLYAFE}_{i,q} + c_6\text{EARLYDISP}_{i,q} + c_7\text{DAMAGE}_{i,q} \\ & + c_8\text{NEGCAR}_{i,q} + c_9\text{NEEDCAP}_{i,q} + e_{i,q} \end{aligned} \tag{3}$$

where $\text{DISC}_{i,q}$ equals 0 if the firm issues no earnings-related disclosures, 1 if the firm issues one disclosure, or 2 if the firm issues multiple disclosures during a quarter. POSTFD_q , as defined previously, is an indicator variable equaling 1 if the firm-quarter is post-FD, and 0 otherwise.

²⁸ Inspection of the data reveals that First Call did not collect this data comprehensively prior to 1995. The number of disclosures reported in the database stabilizes from approximately 1995 onward. Accordingly, we compute the Z-statistics reported in Tables 5 and 6 using only data from the first quarter of 1995 through the second quarter of 2000.

²⁹ Kile et al. (1998) report that between 39 and 53 percent of their sample make at least one voluntary disclosure per year. Approximately 45 percent of our sample firms make at least one voluntary disclosure in 2000 and approximately 55 percent make at least one voluntary disclosure in 2001, which is comparable to that reported by Kile et al.(1998). The frequencies we report in Table 5 are by quarter and are thus lower.

³⁰ Testing whether firms issuing disclosures in the pre-FD period issue more disclosures after FD is not practical because most firms do not issue disclosures in consecutive like quarters. Thus, if a firm discloses in a particular quarter, the odds are high that the firm will not disclose again in the like quarter next year. For this reason, we classify firms as "disclosing firms" if they disclose in either our pre- or post-FD quarters.

³¹ Tests based on the means of the within-firm pre- to post-FD changes produce results very similar to those reported in Table 5.

Theoretical and empirical research suggests firms face incentives to issue disclosures to reduce information asymmetry between managers and various stakeholders. For example, the “expectation adjustment hypothesis” (Ajinkya and Gift 1984) suggests managers voluntarily disclose information when market expectations differ substantially from management’s beliefs. We include several variables reflecting, albeit imperfectly, measures of information asymmetry between managers and investors in an attempt to control for the portion of information asymmetry not due to FD. Specifically, we include the absolute cumulative abnormal return over the quarter ($ABSCAR_{i,q}$), as larger price movements suggest greater information flows and, therefore, greater potential for managers to have information investors do not. We include the absolute deviation of the firms’ earnings-price ratio from the industry median at the end of the fiscal quarter ($EPDEV_{i,q}$), since price-earnings multiples outside industry norms increase the likelihood investors’ valuations differ from management’s. We include the mean absolute analyst forecast error ($EARLYAFE_{i,q}$) and forecast dispersion ($EARLYDISP_{i,q}$), both measured on the day after the previous quarter’s earnings announcement, since larger values of these variables suggest inaccurate and diverse initial expectations of earnings, increasing the likelihood that the market’s expectations deviate from management’s.

Research suggests managers have incentives to disclose when expected litigation costs are high (Skinner 1997). Following Skinner (1997), we measure potential damages from litigation as the dollar amount of investors’ losses, abstracting from the change in the market index, from trading the firm’s stock over the period under consideration. Specifically, $DAMAGE_{i,q}$ equals $-(MVE_{i,q})(MKTADJRET_{i,q})(1 - (1 - VOL_{i,q})^{N_{i,q}})$ where $MVE_{i,q}$ is the market value of equity 10 days before the first disclosure (10 days before the earnings announcement for firms without voluntary disclosures), $MKTADJRET_{i,q}$ is market-adjusted return, and $VOL_{i,q}$ is the average daily percentage of the firm’s shares traded, both measured over the $N_{i,q}$ day period between the current and previous earnings announcement. As in Skinner (1997), we set negative values of $DAMAGE_{i,q}$ equal to \$1 million. We also include an indicator variable for negative returns ($NEGCAR_{i,q}$) as an additional proxy for litigation risk. Finally, to control for the *ex ante* need for capital, we include $NEEDCAP_{i,q}$, defined as operating cash flows less capital expenditures, scaled by current assets (Dechow et al. 1996).

Panel A of Table 6 reports descriptive data regarding the control variables. The mean quarterly absolute cumulative abnormal return ($ABSCAR_{i,q}$) is 0.2705 and cumulative abnormal returns are negative ($NEGCAR_{i,q}$) for 53.03 percent of the firm-quarters. Earnings-to-price ratios deviate, on average, from the industry median at the end of the quarter ($EPDEV_{i,q}$) by 0.55 percent of stock price. Forecast errors ($EARLYAFE_{i,q}$) and forecast dispersion ($EARLYDISP_{i,q}$) early in the quarter average 0.58 and 0.21 percent of stock price, respectively. $DAMAGE_{i,q}$ averages 97.39, indicating that average potential litigation damages, for the typical firm-quarter in our sample, are just shy of \$100 million. $NEEDCAP_{i,q}$ is negative, on average, suggesting the average firm in our sample needs external capital during our sample period (i.e., capital expenditures exceed operating cash flows).

The pre-FD means and medians are statistically different from their post-FD counterparts for all control variables. For our sample, absolute cumulative abnormal returns, the absolute difference between firm-specific and industry median earnings to price ratios, expected litigation damages, the percentage of sample firms with negative cumulative abnormal returns, and the need for external capital are all lower post-FD than pre-FD. Nonetheless, analysts’ forecast errors and forecast dispersion, measured early in the quarter, are both higher in our post-FD period. Panel B of Table 6 reveals that most of the

TABLE 6
Ordered-Logit Regression of Voluntary Forward-Looking Earnings-Related Disclosure Occurrences
on Post-FD Indicator and Control Variables

$$DISC_{i,q} = c_0 + c_1 + c_2POSTFD_q + c_3ABSCAR_{i,q} + c_4EPDEV_{i,q} + c_5EARLYAFE_{i,q} + c_6EARLYDISP_{i,q} + c_7DAMAGE_{i,q} + c_8NEGCAR_{i,q} + c_9NEEDCAP_{i,q} + e_{i,q} \tag{3}$$

Panel A: Distributional Statistics

Control Variables ^a	Means			p-value ^b	Medians			p-value
	Overall	Pre-FD	Post-Fd		Overall	Pre-FD	Post-FD	
ABSCAR	0.2705	0.2823	0.2587	(0.00)	0.1924	0.1975	0.1854	(0.02)
EPDEV	0.0055	0.0056	0.0054	(0.10)	0.0028	0.0030	0.0026	(0.00)
EARLYAFE	0.0058	0.0052	0.0063	(0.00)	0.0018	0.0017	0.0020	(0.00)
EARLYDISP	0.0021	0.0020	0.0022	(0.00)	0.0007	0.0007	0.0008	(0.00)
DAMAGE	97.3927	121.0256	73.7600	(0.00)	1.0000	1.0559	1.0000	(0.00)
NEGCAR	0.5303	0.5589	0.5017	(0.00)	1.0000	1.0000	1.0000	(0.00)
NEEDCAP	-0.0323	-0.0389	-0.0256	(0.09)	0.0050	0.0020	0.0082	(0.03)

Panel B: Pearson Correlations^c

	ABSCAR	EPDEV	EARLYAFE	EARLYDISP	DAMAGE	NEGCAR	NEEDCAP
POSTFD	-0.0364	-0.0255	0.0523	0.0396	-0.0642	-0.0561	0.0185
ABSCAR		-0.0090	0.0700	0.0449	0.0304	-0.1425	-0.1187
EPDEV			0.1814	0.1468	-0.0405	-0.0392	0.0638
EARLYAFE				0.4417	-0.0680	0.0137	-0.1096
EARLYDISP					-0.0640	0.0074	-0.0952
DAMAGE						0.2248	0.0310
NEEDCAP							-0.0241

(continued on next page)

TABLE 6 (continued)

Panel C: Ordered-Logit Results		Disclosures about:					
		Current Earnings ^d		Future Earnings		Total	
		Coefficient	p-value	Coefficient	p-value	Coefficient	p-value
Variable							
Intercept 1		-4.336	(0.00)	-3.408	(0.00)	-2.716	(0.00)
Intercept 2		-2.138	(0.00)	-2.337	(0.00)	-1.664	(0.00)
POSTFD		0.819	(0.00)	1.380	(0.00)	1.134	(0.00)
ABSCAR		-0.060	(0.60)	-0.403	(0.00)	-0.200	(0.06)
EPDEV		-0.446	(0.92)	10.158	(0.03)	5.359	(0.15)
EARLYAFE		28.844	(0.00)	-8.837	(0.03)	16.844	(0.00)
EARLYDISP		-46.943	(0.00)	-48.062	(0.00)	-52.264	(0.00)
DAMAGE ^e		0.557	(0.00)	0.690	(0.00)	0.672	(0.00)
NEGCAR		0.138	(0.04)	-0.103	(0.14)	0.007	(0.91)
NEEDCAP		0.527	(0.00)	0.588	(0.00)	0.589	(0.00)
Pseudo R ² %		4.98		7.11		8.10	
Variable		Z-Statistic ^f	p-value	Z-Statistic	p-value	Z-Statistic	p-value
POSTFD		2.748	(0.00)	2.832	(0.00)	3.489	(0.00)

(continued on next page)

TABLE 6 (continued)

^a Variable definitions: $DISC_{i,q} = 0$ if the firm issues no voluntary earnings-related disclosures, 1 if the firm issues one disclosure, or 2 if the firm issues multiple disclosures during a quarter; $POSTFD_q = 1$ if the observation is from the fourth quarter of 2000 or the first or second quarters of 2001, and 0 otherwise; $ABSCAR_{i,q}$ is the absolute value of the cumulative abnormal return from 64 days before through two days after the earnings announcement; $EPDEV_{i,q} =$ the absolute deviation of the firms' earnings-price ratio from the industry median at the end of the fiscal quarter; $EARLYAFE_{i,q} =$ the absolute mean forecast error as of the day after the previous quarter's earnings announcement; $EARLYDISP_{i,q} =$ the standard deviation in individual analysts' forecasts as of the same date as $EARLYAFE_{i,q}$; $DAMAGE_{i,q} =$ expected litigation damages computed as in Skinner (1997) in \$ millions; $NEGCAR_{i,q} = 1$ if the cumulative abnormal return from 64 days before through two days after the earnings announcement is negative, and 0 otherwise; $NEEDCAP_{i,q} =$ expected need for capital, computed as in Dechow et al. (1996). We winzorize all continuous variables at the 99th percentile of the distributions of their absolute values.
^b All p-values are two sided. p-values for means are from t-tests of the difference between the pre- and post-FD means. For medians, p-values are from Wilcoxon two-sample tests. p-values for ordered-logit parameter estimates are based on the Wald test. p-values for Z-statistics are from the student's t-distribution. There are 7,064 observations.
^c All correlations are significant at the 0.05 level (two-sided) or better except for the correlation between $NEGCAR_{i,q}$ and $NEEDCAP_{i,q}$, which is significant at the 0.06 level (two-sided), and the correlations between $NEGCAR_{i,q}$ and $EARLYAFE_{i,q}$, $NEGCAR_{i,q}$ and $EARLYDISP_{i,q}$, $ABSCAR_{i,q}$ and $EPDEV_{i,q}$, and $POSTFD_q$ and $NEEDCAP_{i,q}$, which are not significant at conventional levels.
^d Current earnings columns refer to voluntary disclosures about the upcoming earnings announcements. Future earnings columns refer to voluntary disclosures about earnings for future periods. The total column includes both types.
^e For presentation purposes, we multiplied this coefficient by 1,000.
^f We estimate the equation for each of the 16 sets of seven consecutive quarters from the first quarter of 1995 through the second quarter of 2000, with $POSTFD_q$ equal to 1 for the last three quarters in each seven-quarter set and 0 for the first three quarters. The middle quarter is discarded. The Z-statistic is equal to $(\hat{c}_2 - \bar{c}_2)/\sigma_{\hat{c}_2}$, where $\hat{c}_2$ is the estimate of c_2 , from the true pre- and post-FD periods and $\bar{c}_2$ is the mean and $\sigma_{\hat{c}_2}$ the standard deviation of the 16 pseudo- $POSTFD_q$ coefficient estimates.

correlations among these variables are statistically significant, although small in magnitude. The most notable exception is the relatively large correlation (0.44) between $\text{EARLYAFE}_{i,q}$ and $\text{EARLYDISP}_{i,q}$. The correlations between the post-FD indicator and the control variables are all less than 0.10.

We estimate Equation (3) separately for total disclosures, disclosures about current earnings, and disclosures about future earnings. Panel C of Table 6 displays the results. The POSTFD_q coefficient is positive and significantly different from zero at better than the 0.01 level (two-sided) in all three regressions. The time series Z-statistic for the POSTFD_q coefficient is also positive and significant at better than the 0.01 level. Thus, after controlling for various non-FD factors potentially influencing firms' voluntary disclosure decisions during our sample period, our evidence suggests firms increased their voluntary, forward-looking, earnings-related disclosures after FD.

The coefficients on our information asymmetry variables are neither consistently significant nor consistently of the expected sign. The coefficient on $\text{ABSCAR}_{i,q}$ is generally negative, contrary to expectations, but is not significant when disclosures about current earnings is the dependent variable. The $\text{EPDEV}_{i,q}$ coefficient is positive and significant only for forecasts of future quarters' earnings, which is sensible since price is a forward-looking variable, i.e., a deviation from the industry earnings-to-price ratio indicates expectations that the firm will perform differently than the industry in future quarters. The $\text{EARLYAFE}_{i,q}$ coefficient is positive and significant for current earnings forecasts alone, which is understandable since $\text{EARLYAFE}_{i,q}$, by design, measures poor expectations of current earnings. The coefficient on $\text{EARLYDISP}_{i,q}$ is negative and significant in all three regressions, inconsistent with the notion that managers make disclosures to reduce heterogeneity in analysts' expectations. However, this result may arise if the number of disclosures is correlated over time, and *past* disclosures reduce current analysts' forecast dispersion. As expected, the coefficient on our litigation risk proxy, $\text{DAMAGE}_{i,q}$, is positive and significant in all models. The coefficient on $\text{NEGCAR}_{i,q}$ is positive and significant when we estimate Equation (3) for disclosures about current earnings, but is not significant otherwise. The coefficient on $\text{NEEDCAP}_{i,q}$ is consistently positive and significant.

We repeat all the univariate and the ordered logit analyses reported in Tables 5 and 6 separately for each of the three sets of like quarters in our sample period with qualitatively similar results in each quarterly comparison. We also estimated Equation (3) including several other variables suggested by prior research, including institutional ownership, number of employees scaled by total assets, number of analysts following the firm, and the natural log of the market value of equity (see Lang and Lundholm 1993, 2002; Healy et al. 1999; Miller and Piotroski 2000; Bushee and Noe 2001; Chen 2002). We also included the book-to-price ratio as another measure of uncertainty about firm value (Barth et al. 2001) and trading volume, as a percentage of outstanding shares, as a measure of heterogeneous beliefs. Results regarding the POSTFD_q coefficient are virtually identical. Estimating Equation (3), after replacing $\text{ABSCAR}_{i,q}$ and $\text{DAMAGE}_{i,q}$ with their natural logs, does not change our inferences.

In summary, our evidence suggests firms increased their voluntary disclosures of forward-looking earnings information after implementation of FD. Our univariate analyses also suggest the proportion of firms making disclosures increased after FD and that disclosing firms increased their disclosure frequency after FD. Our results are robust to controls for alternative non-FD motivations for voluntary disclosures, and are unusual relative to time-series variation.

VI. CONCLUSION

Critics of FD suggest private communication to analysts is an important source of material information to the capital markets. They argue that, by essentially prohibiting private communication between firms and analysts, FD will impair the average level of information in the capital markets, resulting in less accurate expectations of firm performance and greater price shocks when performance is revealed. In this paper, we analyze the effect of FD on the financial information environment prior to a key source of information to the capital markets: quarterly earnings announcements. We find absolute cumulative abnormal returns prior to earnings announcements are significantly smaller post-FD than pre-FD. These decreases are significant compared to random variation in absolute cumulative abnormal returns over time and robust to controls for other factors affecting the earnings-returns relation and suggest stock prices incorporate earnings-related information earlier after FD's implementation. Simple univariate tests on mean and median absolute forecast errors and forecast dispersion suggest analysts' forecasting ability declined after FD's implementation. However, these impairments are not statistically significant relative to the historical time-series of changes in forecast errors and dispersion, and we find no significant reduction in analysts' performance after FD when we control for non-FD determinants of analysts' forecast accuracy and dispersion. Finally, we find a marked increase in the frequency of voluntary forward-looking disclosures by firms after FD became effective, even after controlling for other factors explaining voluntary disclosure.

In summary, other than mixed and weak univariate evidence on analysts' forecast performance, we find no reliable evidence of significant deterioration in the information environment prior to earnings announcements, after implementation of FD. On the contrary, some of our analyses suggest an improvement. For example, we find improved information efficiency of stock prices and a substantial increase in the frequency with which firms voluntarily disclose forward-looking earnings-related information. Our evidence that firms issue more voluntary, earnings-related disclosures is consistent with firms substituting voluntary, public disclosures to at least partially offset reductions in information flows through analysts.

While we have attempted to control for other non-FD factors that could affect our inferences, since our event (implementation of FD) occurs simultaneously for all firms, we can never completely rule out the possibility that our results are attributable to some other unknown contemporaneous economic event unrelated to FD. Also, our study documents *average* effects. We leave to future research examinations of cross-sectional differences in the effect of Regulation FD. Finally, our study investigates the immediate short-term effects of FD. While evidence on the immediate effect of FD is of interest from a policy perspective, the long-term effects of FD are unknown. Thus, although our results provide interesting early evidence on the short-term effect of Regulation FD, the study's results should be interpreted with caution.

REFERENCES

- Ajinkya, B., and M. Gift. 1984. Corporate managers' earnings forecasts and symmetrical adjustments of market expectations. *Journal of Accounting Research* 22 (Autumn): 425-444.
- , R. Atiase, and M. Gift. 1991. Volume of trading and the dispersion in financial analysts' forecasts. *The Accounting Review* 66 (April): 389-401.
- Association for Investment Management and Research (AIMR). 2001. FD e-Survey summary. Retrieved June 2001 from http://www.aimr.com/pressroom/01releases/regfd_surveysum.htm.
- Atiase, R. K. 1985. Predisclosure information, firm capitalization, and security price behavior around earnings announcements. *Journal of Accounting Research* 23 (Spring): 21-36.

- Ball, R., and P. Brown. 1968. An empirical evaluation of accounting income numbers. *Journal of Accounting Research* 6 (Autumn): 159–178.
- , and E. Bartov. 1996. How naive is the stock market's use of accounting information? *Journal of Accounting and Economics* 21 (June): 319–337.
- Bamber, L. S., T. E. Christensen, and K. M. Gaver. 2000. Do we really “know” what we think we know? A case study of seminal research and its subsequent overgeneralization. *Accounting, Organizations and Society* 25 (February): 103–129.
- Barth, M., R. Kasznik, and M. McNichols. 2001. Analyst coverage and intangible assets. *Journal of Accounting Research* 39 (June): 1–34.
- Beaver, W. H. 1968. The information content of annual earnings announcements. *Journal of Accounting Research* 6 (Supplement): 67–92.
- , R. Lambert, and D. Morse. 1980. The information content of security prices. *Journal of Accounting and Economics* 2 (March): 3–28.
- Bernard, V. 1987. Cross-sectional dependence and problems in inference in market-based accounting research. *Journal of Accounting Research* 25 (Spring): 1–48.
- , and J. Thomas. 1989. Post-earnings-announcement drift: Delayed price response or risk premium. *Journal of Accounting Research* 27 (Supplement): 1–36.
- , and ———. 1990. Evidence that stock prices do not fully reflect the implications of current earnings for future earnings. *Journal of Accounting and Economics* 13 (December): 305–340.
- Black, F. 1976. Studies of stock price volatility changes. In *Proceedings of the 1976 Meetings of the American Statistical Association, Business and Economics Section*, 177–181.
- Brown, L. D., R. L. Hagerman, P. A. Griffin, and M. E. Zmijewski. 1987. Security analyst superiority relative to univariate time series models in forecasting quarterly earnings. *Journal of Accounting and Economics* 9 (April): 61–88.
- Bushee, B., and C. Noe. 2001. Corporate disclosure practices, institutional investors, and stock return volatility. *Journal of Accounting Research* 38 (Supplement): 207–246.
- Chaney, P., C. Hogan, and D. Jeter. 1999. The effect of reporting restructuring charges on analysts' forecast revisions and errors. *Journal of Accounting and Economics* 27 (June): 261–284.
- Chen, S. 2002. Managerial guidance of market expectations: Incentives and effects. Working paper, University of Southern California, Los Angeles.
- Cheon, Y. S., T. E. Christensen, and L. S. Bamber. 2001. Factors associated with differences in the magnitude of abnormal returns around NYSE vs. NASDAQ firms' earnings announcements. *Journal of Business Finance and Accounting* 28 (November): 1073–1108.
- Christie, A. A. 1982. The stochastic behavior of common stock variances: Value, leverage and interest rate effects. *Journal of Financial Economics* 10 (December): 407–432.
- Clement, M. 1999. Analyst forecast accuracy: Do ability, resources, and portfolio complexity matter? *Journal of Accounting and Economics* 27 (June): 285–303.
- Collins, D., and S. P. Kothari. 1989. An analysis of the intertemporal and cross-sectional determinants of earnings response coefficients. *Journal of Accounting and Economics* 11 (July): 143–181.
- Dechow, P. M., R. G. Sloan, and A. P. Sweeney. 1996. Causes and consequences of earnings manipulations: An analysis of firms subject to enforcement actions by the SEC. *Contemporary Accounting Research* 13 (Spring): 1–36.
- Dempsey, S. J. 1989. Predisclosure information search incentives, analyst following, and earnings announcement price response. *The Accounting Review* 64 (October): 748–757.
- Diamond, D. W., and R. E. Verrecchia. 1991. Disclosure, liquidity, and the cost of capital. *Journal of Finance* 46 (September): 1325–1359.
- Francis, J., D. Hanna, and L. Vincent. 1996. Causes and effects of discretionary asset writeoffs. *Journal of Accounting Research* 34 (Supplement): 117–134.
- , K. Schipper, and L. Vincent. 2001. Earnings announcements and competing information. Working paper, Duke University, Durham, NC.
- , ———, and ———. 2002. Expanded disclosures and the increased usefulness of earnings announcements. *The Accounting Review* 77 (July): 515–546.

- Freeman, R. N. 1987. The association between accounting earnings and security returns for large and small firms. *Journal of Accounting and Economics* 9 (July): 195–228.
- , and S. Tse. 1989. The multi-period information content of accounting earnings: Confirmations and contradictions of previous earnings reports. *Journal of Accounting Research* 27 (Supplement): 49–79.
- Grant, E. B. 1980. Market implications of differential amounts of interim information. *Journal of Accounting Research* (Spring): 255–268.
- Greene, W. H. 1997. *Econometric Analysis*. Upper Saddle River, NJ: Prentice Hall.
- Hassett, K. A. 2000. Outlaw selective disclosure?—No, the more information the better. *Wall Street Journal* (August 10): A18.
- Hayn, C. 1995. The information content of losses. *Journal of Accounting and Economics* 29 (September): 125–153.
- Healy, P., A. Hutton, and K. Palepu. 1999. Stock performance and intermediation changes surrounding sustained increases in disclosure. *Contemporary Accounting Research* 16 (Fall): 485–520.
- Holthausen, R. W., and R. E. Verrecchia. 1988. The effect of sequential information releases on the variance of price changes in an intertemporal multi-asset market. *Journal of Accounting Research* 26 (Spring): 82–106.
- Jacob, J., T. Lys, and M. Neale. 1999. Expertise in forecasting performance of security analysts. *Journal of Accounting and Economics* 28 (November): 51–82.
- , ———, and J. Sabino. 2000. Autocorrelation structure of forecast errors from time-series models: Alternative assessments of the causes of post-earnings announcement drift. *Journal of Accounting and Economics* 28 (December): 329–358.
- Kile, C., G. Pownall, and G. Waymire. 1998. How frequently do managers disclose prospective earnings information? *Journal of Financial Statement Analysis* 3 (Spring): 5–16.
- Kim, M. S., and W. Kross. 2002. Differences between market responses to earnings announcements in the 1990s vs. 1960s. Working paper, University of Missouri, Columbia, MO.
- Kross, W., and D. Schroeder. 1989. Firm prominence and the differential information content of quarterly earnings announcements. *Journal of Business, Finance, and Accounting* 16 (Spring): 55–74.
- , B. Ro, and D. Schroeder. 1990. Earnings expectations: The analysts' information advantage. *The Accounting Review* 65 (April): 461–476.
- Landsman, W., and E. Maydew. 2002. Has the information content of annual earnings announcements declined in the past three decades? *Journal of Accounting Research* 40 (June): 797–808.
- Lang, M. H., and R. J. Lundholm. 1993. Cross-sectional determinants of analyst ratings of corporate disclosures. *Journal of Accounting Research* 31 (Autumn): 246–271.
- , and ———. 1996. Corporate disclosure policy and analyst behavior. *The Accounting Review* 71 (October): 467–492.
- , and ———. 2000. Voluntary disclosure and equity offerings: Reducing information asymmetry or hyping the stock? *Contemporary Accounting Research* 17 (Winter): 623–662.
- Lim, T. 2001. Rationality and analysts' forecast bias. *The Journal of Finance* 56 (February): 369–385.
- Lo, A. W., and A. C. MacKinlay. 1988. Stock market prices do not follow random walks: Evidence from a simple specification test. *Review of Financial Studies* 1 (Spring): 41–66.
- Mendenhall, R., and W. Nichols. 1988. Bad news and differential market reactions to announcements of earlier-quarters vs. fourth-quarter earnings. *Journal of Accounting Research* 26 (Spring): 63–90.
- , and D. Fehrs. 1999. Option listing and the stock price response to earnings announcements. *Journal of Accounting and Economics* 27 (February): 57–87.
- Miller, G., and J. Piotroski. 2000. Forward looking earnings statements: Determinants and market response. Working paper, Harvard University, Cambridge, MA.
- Mohanram, P., and S. Sunder. 2002. Has regulation fair disclosure affected financial analysts' ability to forecast earnings? Working paper, New York University, New York, NY.

- Morse, D. 1981. Price and trading volume reaction surrounding earnings announcements: A closer examination. *Journal of Accounting Research* 19 (Autumn): 374–398.
- Opdyke, J. D., and A. Lucchetti. 2000. Mum's the word in wake of disclosure rule. *Wall Street Journal* (August 16): C1.
- Parks, R. W. 1967. Efficient estimation of a system of regression equations when disturbances are both serially and contemporaneously correlated. *Journal of the American Statistical Association* 62 (June): 500–509.
- Penman, S. 1996. The articulation of price-earnings ratios and market-to-book ratios and the evaluation of growth. *Journal of Accounting Research* 34 (Autumn): 235–259.
- Pulliam, S. 1999. Abercrombie & Fitch ignites controversy over possible leak of sluggish sales data. *Wall Street Journal* (October 14): C1.
- Securities Industry Association (SIA). 2001. Costs and benefits of regulation fair disclosure. New York, NY: SIA.
- Shane, P., N. Soderstrom, and S. W. Yoon. 2001. Earnings and price discovery in the post-reg. FD information environment: A preliminary analysis. Working paper, University of Colorado at Boulder.
- Skinner, D. 1997. Earnings disclosures and stockholder lawsuits. *Journal of Accounting and Economics* 23 (November): 249–282.
- Weber, J. 2000. Full disclosure for all the web allows corporations to share information quickly and cheaply. *Business Week* (September 18): 106.
- Williams, M., and R. McGough. 2000. Intel's jolt shows shifts in market's dynamics—New disclosure rules mean more legwork ahead for analysts. *Wall Street Journal* (September 25): C1.
- Wild, J. J., L. A. Bernstein, and K. R. Subramanyam. 2001. *Financial Statement Analysis*. Seventh edition. New York, NY: McGraw-Hill/Irwin.
- Ziebart, D. 1990. The association between consensus of beliefs and trading activity surrounding earnings announcements. *The Accounting Review* 65 (April): 477–488.